



ros2_control on the real robot

Using ros2_control to drive our robot (off the edge of the bench...)



In the [last tutorial](#) we used ros2_control to drive our robot in Gazebo, but this time we're going to step things up and drive our robot in the real world. It won't make much sense if you haven't read the last tutorial, so make sure you do it first!

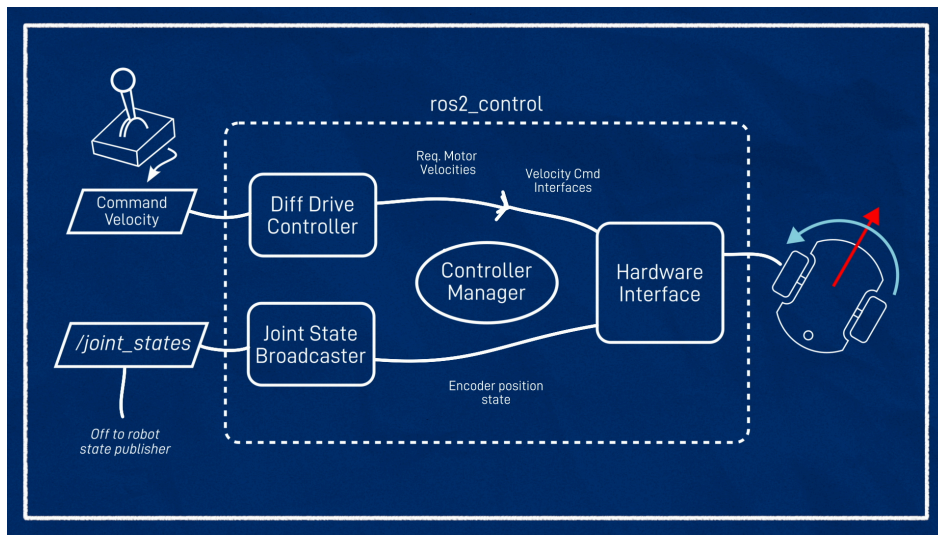
Before we dive into the code, let's quickly recap how ros2_control will work with our robot. At one end, we have a body velocity request, that's the velocity we want the robot to be moving through space, and we call this the *command velocity*. This could come from an operator, or something like a navigation stack. At the other end, we've got the physical robot, with two drive wheels, each of which are velocity controlled.

Although the command velocity is going to be a ROS topic of type `Twist` (or `TwistStamped` as we'll see in the next video), that is a 6-dimensional velocity but since this particular robot's motion is constrained to move forwards and backwards in X, and rotate around Z, we'll only use two of the values (`linear.x` and `angular.z`).



The tool we're going to use to link up this command velocity to the actual motor velocities is ros2_control. This little ecosystem consists of three main parts:

- The `diff_drive_controller` plugin, provided by `ros2_controllers`, which turns our command velocity into abstract wheel velocities
- The hardware interface plugin, provided by us, which turns abstract wheel velocities into signals for the motor controller (e.g. via serial commands)
- The controller manager, provided by `ros2_control`, in this case using the `ros2_control_node`, which links everything together



In addition to this, we have a controller called a joint state broadcaster. All this one does is read the motor encoder positions (provided by the hardware interface) and publish them to the `/joint_states` topic for robot state publisher to generate the wheel transforms. For this kind of robot it's not strictly necessary but it makes the system "complete".

In the previous tutorial we got all this working with a simulated robot, and a Gazebo-integrated controller manager, but this time we'll be doing it with a regular controller manager, our own hardware interface, on a real robot!

Getting the hardware interface

For this to work, we need a hardware interface for our robot. I've written one that you can use if you are using the same Arduino motor control code that I am, otherwise you'll need to find one online or write your own. [Here's an example](#) of one I found (have not tested) for Dynamixel motors.

Writing your own hardware interface is out of the scope of this tutorial, but if people are interested let me know, because they changed the API after foxy, and so I need to rewrite mine for people on newer distros, and that might be a good opportunity to talk through it.

My hardware interface is called `diffdrive_arduino`, and all it does is expose the two velocity command interfaces (one for each motor), and the velocity and position state interfaces. Underneath, it's simply just sending those "m" and "e" commands over serial that we saw in the motor tutorial (*which never got a written version...oops!*). Remember, the `diff_drive_controller` will be handling the actual control allocation.

It would be nice to be able to just `sudo apt install diffdrive_arduino` but unfortunately we can't do that just yet. Right now my driver depends on a great serial library which was never officially updated and released for ROS 2. There is a source port though, so we're going to have to add it to our workspace and compile it ourselves. At some point I might rewrite my code to use a different library which IS available, but I did like that one so William or Scott if you're listening, please consider a ros 2 release. If that ever happens or I rewrite it, I'll release mine as well so that you can install it all through apt.

To get the hardware interface we'll clone from my GitHub into our workspace and build it from source. It also needs a copy of the serial library it depends on, and to keep things simple I've made my own fork (of a different fork). So let's hop onto the Pi and clone those into our workspace, then build them.

```
cd robot_ws/src
git clone https://github.com/joshnewans/serial
git clone https://github.com/joshnewans/diffdrive_arduino
cd ..
colcon build --symlink-install
```

Remember if you haven't cloned your main project package (e.g. `articubot_one` for me) onto the robot yet to do that one as well, and if you *have* already cloned it you'll want to pull the latest changes (from the last tutorial) from git.

Update the URDF

Now we need to update our URDF file to reflect the new hardware interface. Remember that we are editing these files **on the Raspberry Pi**.

TIP

To edit files on the Pi I like to use VS Code Remote Development. You can also just do it in the terminal with your favourite command line editor, or on your development machine and sync the files across to test them.

Let's open up the `ros2_control1.xacro` file we made in the last video. Here we will find the `<ros2_control1>` block we wrote in the last tutorial to describe our hardware interface when we were using Gazebo, but now we need one for the real system.

The steps to do that are:

- Make a copy of the Gazebo one and comment out the original. We'll come back to it later so we can toggle between them.
- Change the name (e.g. to `"RealRobot"`).
- Change the plugin to `diffdrive_arduino/DiffDriveArduino`
- Add the parameters for the hardware interface.

Hardware interface plugins can expose a bunch of parameters for us to set. In this case we'll have the following:

- Left and right wheel joint names

- Arduino loop rate
- Serial port
- Baud rate
- Comms timeout
- Encoder counts per revolution

You'll need to set the parameters appropriately for your system. The result should look something like:

```
<ros2_control name="RealRobot" type="system">
  <hardware>
    <plugin>diffdrive_arduino/DiffDriveArduino</plugin>
    <param name="left_wheel_name">left_wheel_joint</param>
    <param name="right_wheel_name">right_wheel_joint</param>
    <param name="loop_rate">30</param>
    <param name="device">/dev/ttyUSB0</param>
    <param name="baud_rate">57600</param>
    <param name="timeout">1000</param>
    <param name="enc_counts_per_rev">3436</param>
  </hardware>
  <!-- Note everything below here is the same as the Gazebo one -->
  <joint name="left_wheel_joint">
    <command_interface name="velocity">
      <param name="min">-10</param>
      <param name="max">10</param>
    </command_interface>
    <state_interface name="velocity"/>
    <state_interface name="position"/>
  </joint>
  <joint name="right_wheel_joint">
    <command_interface name="velocity">
      <param name="min">-10</param>
      <param name="max">10</param>
    </command_interface>
    <state_interface name="velocity"/>
    <state_interface name="position"/>
  </joint>
</ros2_control>
```

Launch file

Now we need a launch file. We'll start by making a copy of `launch_sim.launch.py` and call it `launch_robot.launch.py`. There are a few things we'll want to change in here.

Add the controller manager node

We'll get rid of the Gazebo launch and spawner since we don't need them any more, but something we will need is a controller manager. Remember that last time the Gazebo plugin was running a controller manager for us, but now we need to start one ourselves. This starts simply enough, we can copy one of our spawners, change the name of the variable, and also change the executable to `ros2_control_node`. We'll also leave a spot for the parameters we're about to fill in.

```
# Delete the Gazebo stuff and replace it with this
controller_manager = Node(
    package='controller_manager',
    executable='ros2_control_node',
    parameters=[XXXXXXXXXX],
)
```

You might recall from the last video that the two things the controller manager needs are the robot description (the URDF), and controller parameters.

Getting the robot description

The robot description we're going to get in a bit of an odd way - it was originally sent as a parameter to `robot_state_publisher` in `rsp.launch.py`, so we need to get it back out. Unfortunately the controller manager can't just read it from the `/robot_description` topic the way the `gazebo_ros2_control` plugin does (though I'm not sure why!).

To do this we'll use a `Command` substitution. This will be evaluated at the time when it goes to run the controller manager, and it will request the contents of the `robot_description` parameter from the `robot_state_publisher` node.

```
robot_description = Command(['ros2 param get --hide-type /robot_state_publisher robot_description'])
```

Getting the controller params

For the controller parameters we just need the path to the params file, and for that we can use the same `os.path.join` function we have in other places.

```
controller_params = os.path.join(
    get_package_share_directory('articubot_one'), # <-- Replace with your package name
    'config',
    'my_controllers.yaml'
)
```

Now we can update that `parameters` line from before, to include our robot description and controller parameters:

```
parameters=[{'robot_description': robot_description},
             controller_params],
```

Don't use sim time

Because we're in the real world now we want to make sure that we set `use_sim_time` to false in both:

- Our launch file `launch_robot.launch.py`, where we include the robot state publisher launch (`rsp.launch.py`)
- Our parameters file `my_controller_params.yaml` (in this case, just remove the line rather than explicitly setting it to false)

Again, this will cause problems for the Gazebo simulation but we'll fix that up later.

Delaying our nodes

One problem we'll run into is that we need the robot state publisher to have finished starting up before we run the `ros2 param get` command, or else we'll fail to start the controller manager and nothing will work. The simplest (though probably not the best) way to do this is to simply add a time delay. The example below uses 3 seconds which should be enough.

```
# At the top with our imports
from launch.actions import IncludeLaunchDescription, TimerAction
...
# Directly below the controller manager node
delayed_controller_manager = TimerAction(period=3.0, actions=[controller_manager])
```

We can also then delay the controller spawners until the controller manager has started. They do have a decent timeout anyway and so will probably be fine, but chaining it together like this is a bit neater.

```
# At the top with our imports
from launch.actions import RegisterEventHandler
from launch.event_handlers import OnProcessStart
...
# Directly below the current spawners
delayed_diff_drive_spawner = RegisterEventHandler(
    event_handler=OnProcessStart(
        target_action=controller_manager,
        on_start=[diff_drive_spawner],
    )
)
delayed_joint_broad_spawner = RegisterEventHandler(
    event_handler=OnProcessStart(
        target_action=controller_manager,
        on_start=[joint_broad_spawner],
    )
)
```

Finally, we want to replace the Gazebo entries with `delayed_controller_manager` in our `LaunchDescription` at the end.

```
return LaunchDescription([
    rsp,
    delayed_controller_manager,
    delayed_diff_drive_spawner,
    delayed_joint_broad_spawner
])
```

Ok, that *should* be all we need. Let's rerun colcon to add the new launch file, and then we can test it!

Testing it out!

Basic driving

WARNING! Make sure you prop your robot up before turning on the controllers for the first time, in case it sends incorrect values to the motors and sends the robot flying!

As a first test, we want to:

- Run our `launch_robot.launch.py` script on the Pi, via SSH
- Start RViz on our development machine, with the fixed frame set to `odom` and various displays enabled (e.g. TF, RobotModel)
- On the dev machine, run `teleop_twist_keyboard` (or your preferred teleop method) with the topics remapped like in the previous tutorial (`-r /cmd_vel:=/diff_cont/cmd_vel_unstamped`)

As we send velocity commands, we should see:

- The wheels spinning in the expected direction
- The RViz display moving as if the robot were actually driving (it doesn't know the wheels are spinning in the air!)
- All the transforms etc. are displayed with no errors

If that seems to be working then it's time to take the robot off its stand and try to drive it around! If you are finding that the wheel speeds are too fast/slow, the on-screen instructions for `teleop_twist_keyboard` show how to adjust them, however we'll look at some better approaches to teleop in the next tutorial.

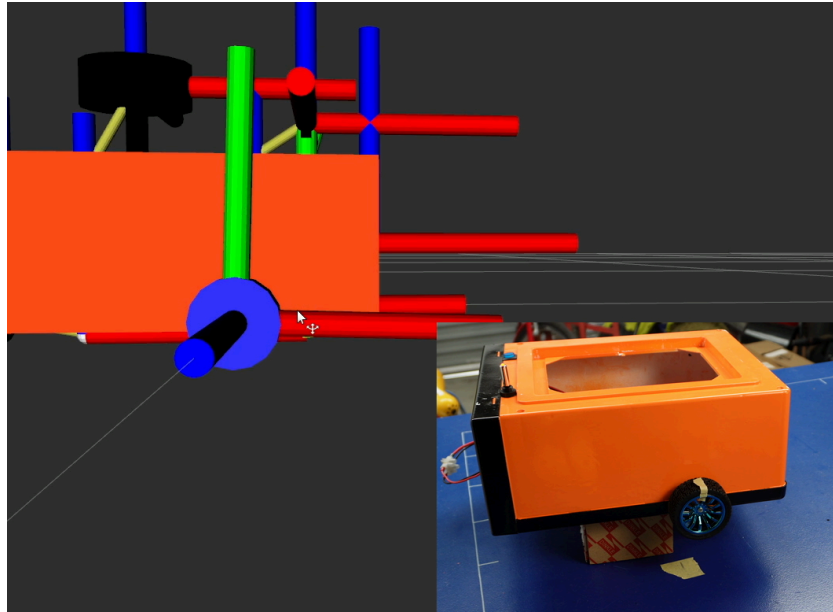
Odometry accuracy

Now we want to see how accurate our odometry is. Odometry will never be perfect, which is why we use other localisation methods such as SLAM or GPS, but it is still worth getting as good as we can. There are three easy steps we can take to assess this (ensure they are done in order as each step is affected by the previous).

1. Encoder accuracy

- Prop the robot back up.
- Restart the controller so that the wheel axes are reset in RViz and note which way they are pointing (e.g. Y is pointing up).
- Set the RViz fixed frame to something robot-fixed (e.g. `base_link`)
- Put a marker (e.g. some tape) on the wheel.
- Drive forward for a while (e.g. 30 revolutions) and stop with the marker in the same spot

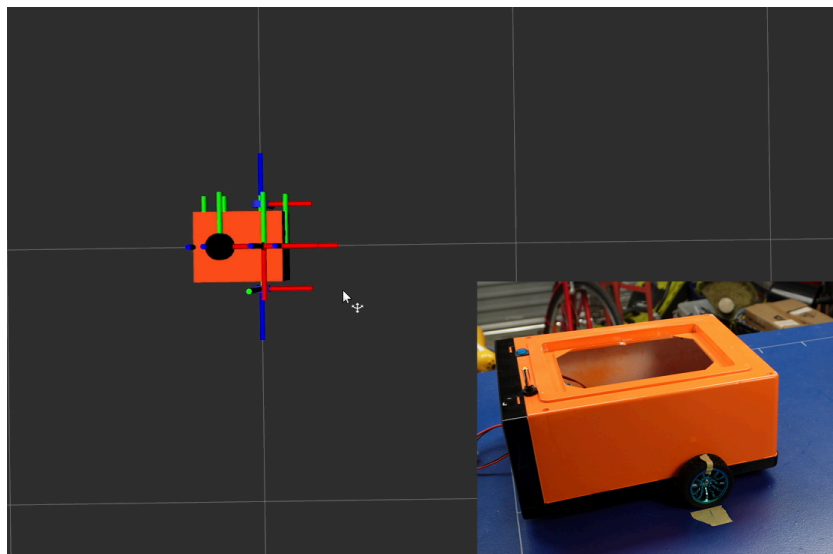
Your axes in RViz should be aligned the same as they were earlier. If it isn't, the encoder counts aren't being properly converted to angular measurements. Check that your counts-per-revolution are correct.



2. Wheel radius

- Place the robot on a large surface and note the wheel locations (e.g. with tape on the ground).
- Make another marker at a known distance ahead (e.g. 1m)
- Restart the controller so that the origin is reset in RViz (fixed frame doesn't matter too much but `odom` is best)
- Drive the robot until the wheels are aligned with the marker

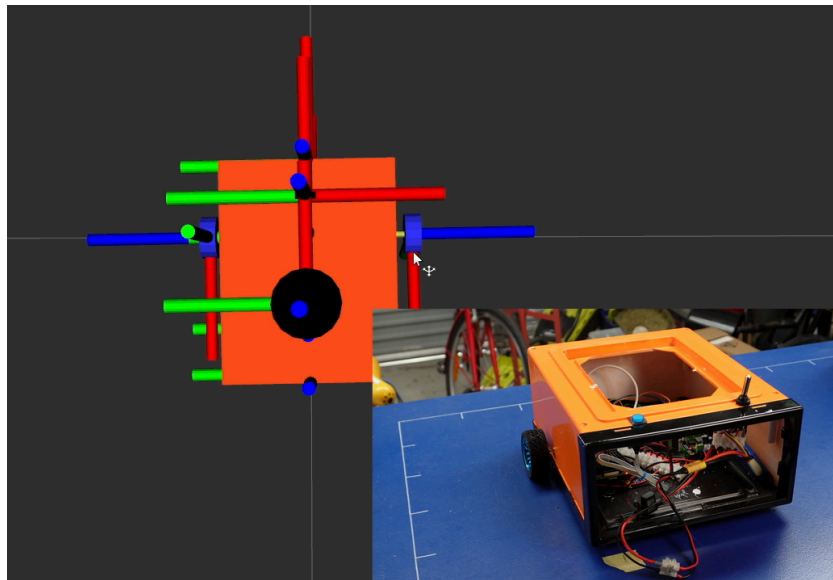
The distance travelled in RViz should be the same as in real life. If not, you need to adjust your wheel radius.



3. Wheel separation

- Place a marker on the ground at the robot wheels
- Restart the controller to reset the robot orientation in RViz
- Drive the robot one revolution so that the wheels are back where they started

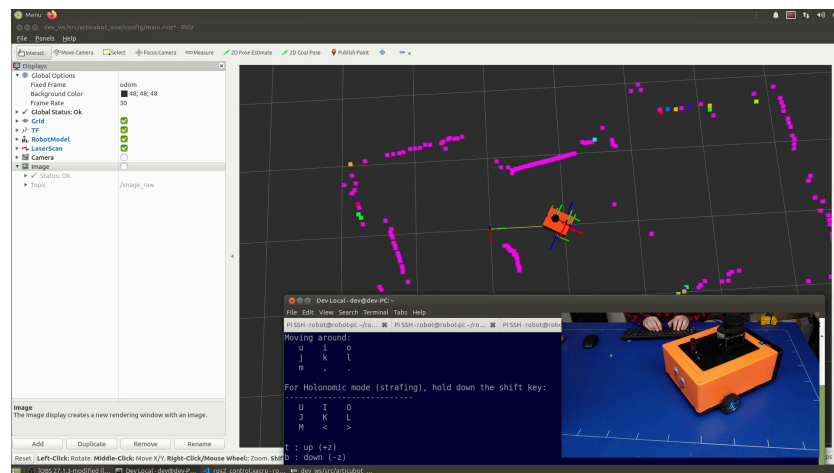
The robot should have turned exactly one revolution in RViz. If not, adjust your wheel separation.



Sensor Integration

This is also a chance to start testing our sensors (lidar, camera, and/or depth camera). We can simply run the [launch files](#) we created for these in previous tutorials.

We should see the laser data come up in RViz, and hopefully as we drive around, the detected objects should stay in approximately the same place. There will be some errors due to timing and also odometry drift.



Unbreaking Gazebo

Now that things are working, let's unbreak it for Gazebo simulations. At this point we should push our changes (on the Pi) to git, and pull them back onto the dev machine.

Xacro switch

The first thing we'll want to do is to put that block we commented out in the xacro onto a toggle. These steps are very similar to what we did in the [extra part](#) of the previous tutorial:

- Add another argument line in our main `robot.urdf.xacro` and call it `sim_mode`, defaulted to `false`.
- In `ros2_control.xacro`, wrap the "real version" in `<xacro:unless value="$(arg sim_mode)">`, and the Gazebo version (uncommented) in `<xacro:if value="$(arg sim_mode)">`.
- In `rsp.launch.py`, expand our xacro command to read as below. Whenever `use_sim_time` is true, we will be enabling `sim_mode`.

```
robot_description_config = Command(['xacro ', xacro_file, ' use_ros2_control:=', use_ros2_control, ' sim_mode:=', use_sim_time])
```

Now, whenever we are simulating, the Gazebo version should be enabled, and when we are running the live system the other one will be. Of course there are many `<gazebo>` tags throughout our URDF that we *could* disable when not in sim mode, but they will not be evaluated anyway so it doesn't really matter.

Controller sim time

Remember that previously we had disabled `use_sim_time` in the `my_controller_params.yaml` file? The problem is that for Gazebo we want this to be turned on. This is a bit messy, and it all comes down to the `gazebo_ros2_control` plugin, so let's swap back to `ros2_control.xacro`.

Ideally, we want to be able to enable `use_sim_time` for the plugin without having to have it inside the params file. Some ways this could possibly work are if the plugin always enabled it (since it's for Gazebo), or had a way to specify single parameters. Right now though, the only option is to specify *multiple parameter files* and it will combine the parameters from all of them.

So we can create a new params file (e.g. `config/gaz_ros2_ctl_sim.yaml`) containing only:

```
controller_manager:
  ros__parameters:
    use_sim_time: true
```

Then, in `ros2_control.xacro` we can add it to the plugin include, so that we have:

```
<gazebo>
  <plugin filename="libgazebo_ros2_control.so" name="gazebo_ros2_control">
    <parameters>$(find articubot_one)/config/my_controllers.yaml</parameters>
    <parameters>$(find articubot_one)/config/gaz_ros2_ctl_sim.yaml</parameters>
  </plugin>
</gazebo>
```

So on the real robot it'll just read the main params file, and on Gazebo it'll do both, and so have sim time enabled.

Note: As of writing this tutorial, this functionality has only just been added and has not been pushed to the package repositories. Until then, you'll need to build `gazebo_ros2_control` yourself by running `git clone -b foxy https://github.com/ros-controls/gazebo_ros2_control` in your dev workspace.

Conclusion

Now we can drive our robot around which is a huge step forward, and hopefully it's starting to feel like a real robot. In the [next tutorial](#) we'll look deeper at *teleoperation* (remote control), including how to control our robot using a game controller, or a smartphone.

**Sasffm**

Oct '22

Hi
i am getting thist error when trying to build diffdrive_arduino with colcon.

```
Starting >>> serial
Starting >>> serial_motor_demo_msgs
Finished <<< serial [1.69s]
Starting >>> diffdrive_arduino
Finished <<< serial_motor_demo_msgs [3.61s]
Starting >>> serial_motor_demo
--- stderr: diffdrive_arduino
CMake Error at CMakeLists.txt:16 (find_package):
  By not providing "Findhardware_interface.cmake" in CMAKE_MODULE_PATH this
  project has asked CMake to find a package configuration file provided by
  "hardware_interface", but CMake did not find one.
```

Could not find a package configuration file provided by
"hardware_interface" with any of the following names:

```
hardware_interfaceConfig.cmake
hardware_interface-config.cmake
```

Add the installation prefix of "hardware_interface" to CMAKE_PREFIX_PATH or
set "hardware_interface_DIR" to a directory containing one of the above
files. If "hardware_interface" provides a separate development package or
SDK, be sure it has been installed.

```
---
Failed <<< diffdrive_arduino [5.06s, exited with code 1]
Aborted <<< serial_motor_demo [3.37s]
```

```
Summary: 2 packages finished [8.30s]
1 package failed: diffdrive_arduino
1 package aborted: serial_motor_demo
1 package had stderr output: diffdrive_arduino
```

does anyone has an idea?
Thanks

[1 reply](#)**JoshNewans**[► Sasffm](#) Oct '22

Did you install ros2_control (e.g. with `sudo apt install ros-foxy-ros2-control`)?
That should provide the hardware_interface dependency.

Remember that if you already installed it on your dev machine but are now building on the Pi you need to install it there too! (I forgot when I was filming haha)

Edit: meant to say there's also another bug preventing it from building which I'm going to fix right now...

[1 reply](#)**Sasffm**

Oct '22

looks like the controller manager services are not available?

[1 reply](#)**JoshNewans**[► Sasffm](#) Oct '22

So just to confirm, did you manage to build diffdrive_arduino after my last reply?

If the services are not available, that suggests the controller manager is not running - you can confirm this with `ros2 node list`. If it is not running, check that it's in your launch file properly, that you are running the correct launch file, and that there are no errors in the output of the launch.

**Sasffm**

Oct '22

The diffdrive_arduino was succesfull build.
But i am getting errors when running "ros2 launch articubot_one launch_robot.launch.py"

```
[INFO] [ros2_control_node-2]: process started with pid [4577]
[INFO] [spawner.py-3]: process started with pid [4579]
[INFO] [spawner.py-4]: process started with pid [4581]
[ros2_control_node-2] [INFO] [1666217546.726356381] [DiffDriveArduino]: Configuring...
[ros2_control_node-2] terminate called after throwing an instance of 'serial::IOException'
[ros2_control_node-2] what(): IO Exception (13): Permission denied, file
/home/rosbot/robot_ws/install/articubot_one/share/articubot_one/config/my_controllers.yaml, line 151.
[spawner.py-4] [INFO] [1666217547.812861551] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-3] [INFO] [1666217547.840100623] [spawner_joint_broad]: Waiting for /controller_manager services
[ERROR] [ros2_control_node-2]: process has died [pid 4577, exit code -6, cmd
'/opt/ros/foxy/lib/controller_manager/ros2_control_node --ros-args --params-file /tmp/launch_params_57dm5z2x --params-
file /home/rosbot/robot_ws/install/articubot_one/share/articubot_one/config/my_controllers.yaml'].
[spawner.py-4] [INFO] [1666217549.844418248] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-3] [INFO] [1666217549.872518979] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-4] [INFO] [1666217551.875385029] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-3] [INFO] [1666217551.904104070] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-4] [INFO] [1666217553.907724653] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-3] [INFO] [1666217553.938034195] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-4] [INFO] [1666217555.937776050] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-3] [INFO] [1666217555.968601738] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-4] [ERROR] [1666217557.765990630] [spawner_diff_cont]: Controller manager not available
[spawner.py-3] [ERROR] [1666217557.798581779] [spawner_joint_broad]: Controller manager not available
[ERROR] [spawner.py-4]: process has died [pid 4581, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py
diff_cont --ros-args'].
[ERROR] [spawner.py-3]: process has died [pid 4579, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py
joint_broad --ros-args'].
```

Why is the controller manager not running.
I use your file from github.

[1 reply](#)**JoshNewans**[► Sasffm](#) Oct '22

Ok so if my guess is correct, this is something I knew I should probably improve in the code which is exception handling to give a more helpful error message.

I think the problem is that it goes to start up the hardware interface but it can't see the device on the specified serial port and so it fails. A few things

to check here:

- Firstly, that you are using the same setup as me (with the Arduino running that same code)
- Have you set the device correctly in `ros2_control.xacro` (e.g. `/dev/ttyUSB0` or `/dev/serial/by-id...`)
- Is the specified device visible in the OS (e.g. if the parameter is set to `/dev/ttyUSB0`, does `ls /dev/ttyUSB0` return something)
- Have you added the user to the `dialout` group with `sudo adduser $USER dialout`?
- Can you connect to the arduino with a serial terminal e.g. `miniterm -e <device path> 57600`
- Has the system got the lidar or other device and the Arduino the wrong way around*
- Did you get the baud rate wrong? (That should be a different kind of problem though)

* I vaguely flagged this in the video, but I should not have used `/dev/ttyUSB0` as that will sometimes be the lidar, instead I should set it by the path or ID (which I will do at some point).

Let me know how you go with that or if you need further assistance 😊

S

Sasffm
Very good hints !
The problem was the 'dialout' group.
After adding the user to the 'dialout' group everything works out fine.
Thank you , that was a excellent support.
Looking forward to the next video .

Oct '22

1 reply

J

JoshNewans
Great - I'm glad we got it sorted!

▶ Sasffm Oct '22

K

ketill
Hello, and thanks for great tutorials. I have noticed that my robot only drives about half the linear and angular velocity as the `/cmd_vel` command. The counts per revolutions is way lower at 392 tics and maybe that could have something to do with the problem? Also its mentioned that the pid uses the unit tics/loop rate. Should I set that number somewhere?

Nov '22

1 reply

J

JoshNewans
Hmm that is strange, but we should be able to figure it out by checking each step along the way. I'm assuming through this that you are using the same Arduino code as me and the `diffdrive_arduino` hardware interface.

▶ ketill Nov '22

- Connect to the Arduino using a serial terminal (e.g. `miniterm`) and confirm your cpr is 392 using the "e" commands
- Calculate the counts per loop to go at 1 rev/sec. In this case that will be $392/30=13$.
- Use the "m" command to set the motors at that speed and confirm it is correct (e.g. with a stopwatch).
- Manually calculate and test the speed that you ultimately want your motors to be going. This might be where you are running into precision issues.
 - e.g. If your `cmd_vel` is 0.5m/s and your wheel radius was 0.06m
 - $0.5/(2 * \pi * 0.06) = 1.32$ rev/sec
 - = 17 counts/loop
 - If your wheels are big and you are trying to drive slow this is where you'll fall apart
 - e.g. if you were aiming for 0.1m/s and had 0.08m radius wheels, you're looking at 2.6 counts/loop
 - This is so low that being off by one will make a huge difference and so the PID controller will have a very hard time tracking it
 - In that case you may want to try driving faster, gearing the motors, lengthining the loop time, or tuning the PID
- Assuming it did track that fine the problem must be further up the chain (which is great, means it's just a software/config issue!)
- Confirm that `ros2_control.xacro` has correct parameters, particularly the loop rate and counts per rev
- Confirm that `my_controllers.yaml` has the correct wheel radius
- At this point it should be working, but if not then you can dig into the `diffdrive_arduino` code (particularly `arduino_comms.cpp` and `diffdrive_arduino.cpp` and print out what that is sending to the Arduino - it should be the same numbers you calculated earlier!
- Maybe also echo the `cmd_vel` topic to verify those numbers are correct.

I hope that helps, please let me know how you go!

1 reply

K

ketill
Thank you for your great and fast answer! And it's amazing to get a great tutorial on the whole motor control chain. For me it was a very hard issue to solve by myself.

▶ JoshNewans Nov '22

I will try all your suggestions. I forgot to mention that the wheel radius is indeed quite large compared with your example (0.09[m]). The wheel separation is 0.43[m]. Also the test of driving a distance and spinning in `rviz2` is correct compared to the real robot. (One meter in `rviz` is one meter on the robot and 360 deg is 360 deg on the robot). I am currently using the full navigation stack so the problem could be higher up but the issue of wrong speed remains when driving with a controller so the issue should not be there (checked agains `/cmd_vel` message).

Don't know where to post this but I found an issue using the Velodyne 16 lidar to laser scan (2D lidar) in Slam toolbox node. It is a quite common lidar so maybe some other have the same issue. It took a long time for me to find the bug but apparently when converting the pointcloud message to laserscan message with the official driver the lidar can give the error of too many or too few scans per revolution. I think if the resolution in the .yaml file is not evenly divided it cases the error and then the slam toolbox is not able to start correctly. I did a bad fix of changing the resolution: 0.006 and it works.

Hope it can be to any help and thanks again. Will update when i have found the issue 😊

D

Douglas_Beach
Hello! I wanted to thank you for these amazing tutorials and ask for some guidance on a problem I just ran into. While trying to build `diffdrive_arduino`, I received the following error:

Dec '22

```
doug@doug-desktop:~/ros2_ws$ colcon build --symlink-install
Starting >>> serial
Starting >>> arduino_com
Starting >>> my_bot
Starting >>> my_robot_controller
Finished <<< my_bot [3.89s]
Starting >>> robot_controller
Finished <<< serial [4.15s]
Starting >>> diffdrive_arduino
--- stderr: diffdrive_arduino
In file included from /home/doug/ros2_ws/src/diffdrive_arduino/src/diffdrive_arduino.cpp:1:
/home/doug/ros2_ws/src/diffdrive_arduino/include/diffdrive_arduino/diffdrive_arduino.h:7:10: fatal error:
hardware_interface/base_interface.hpp: No such file or directory
   7 | #include "hardware_interface/base_interface.hpp"
     | ~~~~~^~~~~~
compilation terminated.
gmake[2]: *** [CMakeFiles/diffdrive_arduino.dir/build.make:76:
CMakeFiles/diffdrive_arduino.dir/src/diffdrive_arduino.cpp.o] Error 1
gmake[1]: *** [CMakeFiles/Makefile2:139: CMakeFiles/diffdrive_arduino.dir/all] Error 2
gmake[1]: *** Waiting for unfinished jobs....
In file included from /home/doug/ros2_ws/src/diffdrive_arduino/src/fake_robot.cpp:1:
/home/doug/ros2_ws/src/diffdrive_arduino/include/diffdrive_arduino/fake_robot.h:7:10: fatal error:
hardware_interface/base_interface.hpp: No such file or directory
   7 | #include "hardware_interface/base_interface.hpp"
     | ~~~~~^~~~~~
compilation terminated.
```

```
gmake[2]: *** [CMakeFiles/take_robot.dir/build.make:70: CMakeFiles/take_robot.dir/src/take_robot.cpp.o] Error 1
gmake[1]: *** [CMakeFiles/Makefile2:165: CMakeFiles/fake_robot.dir/all] Error 2
gmake: *** [Makefile:146: all] Error 2

---
Failed    <<< diffdrive_arduino [5.76s, exited with code 2]
Aborted   <<< my_robot_controller [10.0s]
Aborted   <<< arduino_com [10.6s]
Aborted   <<< robot_controller [8.27s]

Summary: 2 packages finished [13.3s]
1 package failed: diffdrive_arduino
3 packages aborted: arduino_com my_robot_controller robot_controller
1 package had stderr output: diffdrive_arduino
```

I think I figured out the problem... I am running ROS Humble and the Ros2_control package for humble doesn't have hardware_interface/base_interface.hpp anymore. https://github.com/ros-controls/ros2_control/tree/humble/hardware_interface/include/hardware_interface

I wanted to check and see if there may be a workaround by editing the diffdrive_arduino.h file to not use base_interface? Any guidance would be greatly appreciated, thanks!

[3 replies](#)

B **BigShark3000** Dec '22

I am using Humble and my HW Plugin is based on the [ros2_control_demos](#) I found that I had to swap "position" and "velocity" name for the state_interfac, in the command_interface, in the ros2_control.xacro file. As the plugin was expecting Position first. This now works for me.

```
<ros2_control name="RealRobot" type="system">
  <hardware>
    <plugin>hover_diffdrive/HoverDiffDrive</plugin>
  </hardware>
  <joint name="left_wheel_joint">
    <command_interface name="velocity">
      <param name="min">-10</param>
      <param name="max">10</param>
    </command_interface>
    <state_interface name="position"/>
    <state_interface name="velocity"/>
  </joint>
  <joint name="right_wheel_joint">
    <command_interface name="velocity">
      <param name="min">-10</param>
      <param name="max">10</param>
    </command_interface>
    <state_interface name="position"/>
    <state_interface name="velocity"/>
  </joint>
</ros2_control>
```

J **JoshNewans** ▶ [Douglas_Beach](#) Dec '22

Hey! This is an issue I'm well aware of (just this week someone [flagged it on GitHub](#) too), but have not put any energy into resolving yet.

As I understand it the fix is a bit more involved than just the dependence on that file, and that it was a more substantial change to the API.

My current plan is that at the end of the build series (which is drawing near!) I will do a video specifically on how to write a hardware interface for humble, and as a demo I will take my foxy one and upgrade it.

In the meantime I'm not sure if anyone else has forked /updated it (I think someone might have but I can't remember). If I do find out I'll post here. As [@BigShark3000](#) helpfully noted, you could have a go at porting it yourself using the ros2_control_demos - if you do then let us know! That's also a helpful note about the requirement to swap the interface order.

[1 reply](#)

D **Douglas_Beach** ▶ [JoshNewans](#) Dec '22

Thanks for the responses BigShark3000 and Josh. I figured it wasn't going to be a simple fix but thought I would ask to be sure! I've been looking through the ros2_control_demos and diffdrive_arduino code to try and understand how it all works... definitely have a ways to go there. If I do manage to get something working, I'll be sure to share.

[1 reply](#)

T **teknetik** Jan '23

This fork seems to work. It compiles I have not actually tested it as such but hopeful:



[1 reply](#)

S **Sudhanshu_ranjan** Jan '23

I am getting this error.

```
ubuntu@ubuntu:~/robot_ws$ colcon build --symlink-install
Starting >>> serial_motor_demo_msgs
--- stderr: serial_motor_demo_msgs
CMake Error at CMakeLists.txt:19 (find_package):
  By not providing "ament_cmake.cmake" in CMAKE_MODULE_PATH this project
  has asked CMake to find a package configuration file provided by
  "ament_cmake", but CMake did not find one.
```

Could not find a package configuration file provided by "ament_cmake" with any of the following names:

```
ament_cmakeConfig.cmake
ament_cmake-config.cmake
```

Add the installation prefix of "ament_cmake" to CMAKE_PREFIX_PATH or set "ament_cmake_DIR" to a directory containing one of the above files. If "ament_cmake" provides a separate development package or SDK, be sure it has been installed.

```
Failed <<< serial_motor_demo_msgs [0.38s, exited with code 1]
```


Summary: 0 packages finished [0.8/s]
1 package failed: serial_motor_demo_msgs
1 package had stderr output: serial_motor_demo_msgs
1 package not processed
ubuntu@ubuntu:~/robot_ws\$

[1 reply](#)



JoshNewans

► [Sudhanshu_ranjan](#) Jan '23

Did you source your ROS installation first? (i.e. `source /opt/ros/foxy/setup.bash`)?

[1 reply](#)



Sudhanshu_ranjan

► [JoshNewans](#) Jan '23

Yes now it work the problem was same that you have mentioned. Thank you



Enno_Stelma

Feb '23

Hi Josh, Thanks for a great series. I got pretty much stuck on getting it all to work on a real robot. I'm running Humble on Ubuntu 22.04. Since ROS-Serial has been replaced by MicroRos I went that route.

Here's my setup:

Ubuntu 22.04 on a virtual machine (some issues along the way as you mentioned but all sorted)
Completed diff drive robot with RPi4 (Ubuntu 22.04), Arduino Due (MicroRos), L298N motor driver.

I can control the robot with Twist messages and I can publish odometry from the Arduino. The problem is getting it to work with `ros2_control` and specifically implementing a hardware interface.

The way I understand it is that `diffdrive_arduino` won't work in this scenario because it relies on ROS-Serial. I did try to use [ros2_control_demos](#) to solve the problem but did not succeed.

So my question is:

Can `diffdrive_arduino` work in this setup?

If so, how do I get that to work?

If `diffdrive_arduino` can't work, what are my options?

The MicroRos and `ros2_control` combo seems pretty new and I can find very little about it. I've been working on this for a few weeks now but I'm not getting any further.

[1 reply](#)



JoshNewans

► [Enno_Stelma](#) Feb '23

Yeah so the confusion here (and it's not just you, others have had the same issue) is that `diffdrive_arduino` does NOT use `rosterial` (or micro-ROS for that matter).

`rosterial` is a method for sending ROS messages over a serial line (to an Arduino or similar). And now with micro-ROS those kinds of problems can be solved differently.

Whereas `diffdrive_arduino` isn't sending ROS messages at all, it is making a pure serial connection to software on the other end, speaking a (very simple) custom protocol. So you shouldn't have `rosterial` or micro-ROS running on the Arduino, it doesn't know anything about ROS, just some simple motor control serial commands. `diffdrive_arduino` is doing all the translation from ROS to serial.

If you *really* want to use micro-ROS on the Arduino, it probably makes sense to skip `ros2_control` completely. And that is a perfectly acceptable setup - just not what I've done in my tutorials. In that case you'll probably want to have the Arduino receiving a `Twist` and handling all the diff-drive calculations (which it seems you are doing), rather than implementing it on the ROS side.

I hope that clears things up, if not please ask further questions!



Enno_Stelma

Feb '23

Thanks for the super quick reply Josh, that clears it all up. I'm going to stick with your route so that I can follow your track in coming videos. I'm going back to the motor control video to implement `RosArduinoBridge` and then implement `diffdrive_arduino`. Plenty of time to play around with MicroRos etc later.



Jon_Peroutka

Feb '23

Hi, I'm having a problem getting my pi to locate the `ros2_control` package using 'sudo apt install'. It just cannot find it. I've updated and upgraded, following the instructions here: [Ubuntu \(binary\) — ROS 2 Documentation: Humble documentation](#)

Any idea what's going wrong? I'm surprised I can't get the system to locate it. Is there a workaround? Thanks for any help!

[1 reply](#)



sampath_ganesh

Feb '23

@[JoshNewans](#) sir . I was able to run the robot in real time but unable to run in the rviz i.e the robot was not moving in the rviz I have tried to change frames but no result , is there any solution I was using `ros2 humble` and 22.04



JoshNewans

► [Jon_Peroutka](#) Mar '23

Hey Jon, this should be fixed with the latest humble update, so if you try an `apt update` and then go again hopefully it will be there. Let me know if you have further troubles.



Praveen_Kumar

Mar '23

Hi,

Thanks for the great tutorial. I followed everything correctly till Ros2 Controller. My setup is a 4 wheel robot instead of 2 wheels. I have used differential drive and setup wheel count as 2. Everything worked perfectly until I added `diffdrive_arduino` package. I have modified the package to take in four wheels as input. However the issues are:

1. When I try driving, all the wheels are reversed. I modified the `diffdrive` package to take inverted inputs (- for all `vel1`, `vel2`, `vel3` and `vel4`). This seems to fix and motors driving correctly
2. Teleop twist keyboard can be used to control the robot only if linear and angular velocity is set to 0.4xxx. Anything more than 5 or less than 3 does not move the wheels at all.
3. Not all motors rotate at the same speed.

So, my questions are:

1. Can I use the previous `serial_motor_demo` library to complete SLAM setup?
2. Any ideas why the robot is not responding to twist correctly?

[1 reply](#)



JoshNewans

Mar '23

Hi Praveen!



Praveen_Kumar:

When I try driving, all the wheels are reversed. I modified the `diffdrive` package to take inverted inputs (- for all `vel1`, `vel2`, `vel3` and `vel4`). This seems to fix and motors driving correctly

There are a bunch of things that wheel reversing can cause depending on your hardware. Getting the wiring right on the driver is the driver out

there are a bunch of places that wheel reversing can occur, depending on your hardware. Getting the wrong pins on the driver in, the driver out, the controller, the wrong axis direction in URDF, etc. It sounds like what you've done there is probably not the "correct" solution, but if it works, it works 😊

 Praveen_Kumar:

Teleop twist keyboard can be used to control the robot only if linear and angular velocity is set to 0.4xxx. Anything more than 5 or less than 3 does not move the wheels at all.

That's very odd. To start with diagnosing I'll need more detail on your motors/controllers/drivers etc. (which might also help us with problem 1). Depending on how things are being mapped it's not unusual for there to be a lower limit that it won't turn below, but an upper limit is strange.

 Praveen_Kumar:

Not all motors rotate at the same speed.

This is unfortunate, but not too surprising. Again, we'll need more details on how you have things hooked up but this is where the PID controller comes into play (which if you're following the tutorials will be external to ROS/Pi stuff, in an Arduino or some other dedicated controller). The quality control of the motors will be a factor, but the PID controller should be able to be tuned to get the motors performing fairly similarly. If you don't have a PID controller and are running in open loop, that might also be the cause of problem 2.

 Praveen_Kumar:

Can I use the previous serial_motor_demo library to complete SLAM setup?

Sort of. You could modify it to accept a Twist and have differential drive control, but you'd be better off fixing the other issues instead and things should just work.

 Praveen_Kumar:

Any ideas why the robot is not responding to twist correctly?

Yeah, by the sound of things it's worth stepping through the details of your setup and making sure the low-level motor control is working before touching the ROS-level stuff. If it's broken then no amount of ROS tweaking will fix things!

So please post some more details about your hardware and how it's wired up (you can upload pictures to imgur and embed them here) and we'll see what we can figure out 😊

 **tiptonnet** Mar '23

[INFO] [launch]: All log files can be found below /home/system/.ros/log/2023-03-21-14-56-32-932971-mapperbot-5148
[INFO] [launch]: Default logging verbosity is set to INFO
[ERROR] [launch]: Caught exception in launch (see debug for traceback): Caught exception when trying to load file of format [py]: name 'Command' is not defined

Is there an import I'm missing?

 **tiptonnet** Mar '23

There most certainly was 😊
from launch.substitutions import Command

 **LeoCleaner** Mar '23

Hi Josh,

Is it possible to set a parameter for defining the Serial Port of arduino in the launch file. I can scan the ports automatically and say on which port is my Lidar and on which is the Arduino Motor Control but I do not know how to set it dynamically in the diffdrive_arduino.

Thank you very much!

Cheers,
Ivo

 **LeoCleaner** May '23

Hi,

I have lidar and arduino motor control on serial ports of my robot (on usb of my raspberry pi). Sometimes lidar and arduino changes serial ports (lidar is on 1 instead of 0 and arduino on 0 instead of 1) and I have to change the configuration of both of them and recompile and start again. Is there an way to make it automatically on the start.

Thank you very much!

Best regards,
Ivo

[1 reply](#)

 **jaronski** May '23

Hi Josh, do you know if it is possible to implement a real controller of a robot (reality) and in parallel a Gazebo simulation so that ros2_control controls both the hardware and the behavior in Gazebo like a digital twin? In your article I understood that its either the real robot or gazebo simulation.

[1 reply](#)

 **Betul_Duran** May '23

Hello

All the hardware and software I use are the same as the one you use. The router is the same. When I connect the router to the development machine with ethernet as you do, and when I connect it to the raspberry pi 4, when I want to control the robot as in the 13th video, it gives a biiiip sound and cuts the internet for a short time. It gives buffer.cpp error, where could I have made a mistake?

[1 reply](#)

 **JoshNewans** [▶ LeoCleaner](#) May '23

Hi Ivo, sorry for not getting back to you (and many other people).

The simplest way to achieve what you want is instead of referencing the devices as /dev/ttyUSB0 or whatever, instead use /dev/serial/by-id/xxx or /dev/serial/by-path/xxx. These paths are tied to either the identifier of the hardware, or the actual USB port it is plugged into, so as long as you keep them in the same ports, the OS should identify them by the same path each time!

If you really want to have it in the launch file you can use an xacro argument and pass it in, similar to how we handle the sim mode and ros2_control mode in rsp.launch.py.

Again, sorry it's late but I hope that helps!

 **JoshNewans** [▶ jaronski](#) May '23

That's a great question and I've never really thought about it!
In short, I think the answer is no...but sort of yes?
The ros2_control controller expects to only control one hardware interface, so to achieve what you're talking about you'd just have to run two independent controllers and have them both listening to the same twist topic (but avoid them publishing to the same feedback)

independent controllers and have them both listening to the same topic (get data from publishing to the same velocity).

You could achieve a similar result if you bypassed `ros2_control` completely and wrote a custom node that interacted with Gazebo and also your robot but it gets...messy (and I suspect probably won't achieve the desired result, especially with something like a mobile robot. It might be good for an arm)



JoshNewans

► Betul_Duran May '23

Wow that is a strange issue, I'm not sure, sorry.

Also I haven't mentioned this but I actually swapped to a different router as I thought the first one was maybe causing me problems...but then I still had problems with the new one too so I never said anything.

When you say it beeps, do you mean the router beeps or the Pi beeps? Perhaps it is a power issue? How are you powering the router?



Ahmed_Tarek

May '23

Hi Josh, thanks a lot for your tutorials,

I am using Arduino UNO instead of Nano, there is no issues when I do launch `launch_robot` file and it tells me that all controllers configured correctly but when I try to drive it with teleop, none of the motors move!!.. my driver EN pins on the UNO are 6,3,9,5.

Please help me, my graduation project is depending on that



Afiq003

Jun '23

Hi Josh, thanks for this great tutorial. However I'm facing issue when unbreaking gazebo. I follow your tutorial exactly, but when i launch my `launch_sim.launch.py`, i get this error instead

```
afiq@afiq-VM:~/dev_ws$ ros2 launch my_bot launch_sim.launch.py world:=src/my_bot/worlds/obstacles.world
[INFO] [launch]: All log files can be found below /home/afiq/.ros/log/2023-06-02-08-25-17-400678-afiq-VM-24746
[INFO] [launch]: Default logging verbosity is set to INFO
Task exception was never retrieved
future: <Task finished name='Task-2' coro=<LaunchService._process_one_event() done, defined at
/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py:226> exception=SubstitutionFailure('executed
command failed. Command: xacro /home/afiq/dev_ws/install/my_bot/share/my_bot/description/robot.urdf.xacro
use_ros2_control:=true sim_mode:=true')>
Traceback (most recent call last):
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 228, in _process_one_event
    await self.__process_event(next_event)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 248, in __process_event
    visit_all_entities_and_collect_futures(entity, self.__context))
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 45, in visit_all_entities_and_collect_futures
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 45, in visit_all_entities_and_collect_futures
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 45, in visit_all_entities_and_collect_futures
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 45, in visit_all_entities_and_collect_futures
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 38, in visit_all_entities_and_collect_futures
    sub_entities = entity.visit(context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/action.py", line 108, in visit
    return self.execute(context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/actions/node.py", line 444, in execute
    self._perform_substitutions(context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/actions/node.py", line 399, in _perform_substitutions
    evaluated_parameters = evaluate_parameters(context, self.__parameters)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/utilities/evaluate_parameters.py", line 164, in
evaluate_parameters
    output_params.append(evaluate_parameter_dict(context, param))
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/utilities/evaluate_parameters.py", line 71, in
evaluate_parameter_dict
    evaluated_value = perform_substitutions(context, list(value))
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/perform_substitutions_impl.py", line 26, in
perform_substitutions
    return ''.join([context.perform_substitution(sub) for sub in subs])
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/perform_substitutions_impl.py", line 26, in
<listcomp>
    return ''.join([context.perform_substitution(sub) for sub in subs])
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_context.py", line 232, in perform_substitution
    return substitution.perform(self)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/substitutions/command.py", line 116, in perform
    raise SubstitutionFailure(f'executed command failed. Command: {command_str}')
launch.substitutions.substitution_failure.SubstitutionFailure: executed command failed. Command: xacro
/home/afiq/dev_ws/install/my_bot/share/my_bot/description/robot.urdf.xacro use_ros2_control:=true sim_mode:=true
```

How do i solve this?

2 replies



Afiq003

► Afiq003 Jun '23

Solved this. Figure out don't need to install gazebo ros2 control package

1 reply



JoshNewans

► Afiq003 Jun '23

I'm glad you got it sorted but I'm a bit confused by your solution.

It seems like xacro itself was failing - possibly you had an error in your URDF?



AddyPete

Jun '23

Hello I'm a beginner in robotics, im mainly a software guy. With that said, i have a question.

1. is it possible that ill just make my own hard code controller instead of using diff drive from ros2 control package? Ill just convert the `cmd_vel` to left and right motor velocity.
2. by doing the mentioned above, what are the disadvantages of this? I'm doing this because i don't know how to get the hardware interfaces of the physical robot, i only know how to send commands which are the left and right motor velocities.

1 reply



JoshNewans

Jun '23

Welcome!

AddyPete:

is it possible that ill just make my own hard code controller instead of using diff drive from ros2 control package? Ill just convert the `cmd_vel` to

left and right motor velocity.

Yep

 AddyPete:

by doing the mentioned above, what are the disadvantages of this? I'm doing this because i don't know how to get the hardware interfaces of the physical robot, i only know how to send commands which are the left and right motor velocities.

The disadvantages are:

- You have to write the code for the differential drive calculations, and you will probably make a mistake/not think of something that the ros2_control developers have thought of. By using their code you know it works and has been tested (at least, most of the time)
- If you ever want to change your hardware out you'll have to modify the whole thing instead of just the hardware interface

When you say

 AddyPete:

i only know how to send commands which are the left and right motor velocities.

What do you mean? Essentially all a ros2_control hardware interface is is a C++ plugin that does that, but takes the input from ros2_control rather than your own calculation.

But yeah, doing it the long way is always an option (and not a bad one for learning how differential drive control works) but at some point it makes sense to take the more robust, flexible option of ros2_control.

(P.S. I've responded to your other message)

[1 reply](#)



AddyPete

[▶ JoshNewans](#) Jun '23

What I mean with i can send only motor commands. Is that by using pyserial and ESP32 in my laptop, i'll just write a certain string that indicated the speed value of both motors for example, the string is "0.5,1.0&\n" (0.5 = left motor speed, 1.0 = right motor speed) then i use arduino.write() to send those command sending it to another ESP32 that has arduino attached to it.

I did this because the hardware guy just told me to send those string via esp32, and he just do the rest of the hardware control thing.

We don't use raspberry pi btw, just like you did.

So it's basically i'm sending a string command with my laptop connected to esp32 then another esp32 connected to arduino just received the command, and i don't know how the hardware guy works around with that string i have sent. I can ask him if necessary to provide more information.

Is our approach in doing this project correct? Is there anyway i can expose state interfaces with robot with our current approach? Thanks

[1 reply](#)



Jeffrey_Jene

Jun '23

I have a sort of weird problem. I've resumed this project after a couple of months and actually started it over as I'd forgotten a bit. The serial motor demo and gui works great on my setup, the motors behave as expected. However, after completing the ros2 control files and code, using teleop causes the motors to go crazy. They will start slowly in the same direction, then they will quickly climb to FULL speed of my motors, one flipping direction in the process and the robot would spin in place at full speed (I have it on blocks so this won't happen). Nothing I've messed with seems to have made any difference. It almost reminds me of an uncontrolled PID controller, like the motors are trying to match a value or catch up with something and max out. Does anyone have any ideas?

[1 reply](#)



JoshNewans

[▶ AddyPete](#) Jun '23

Honestly, that sounds very similar to how mine works, just with a few extra steps - which means it shouldn't be too hard to modify! I recommend watching my [motor tutorial](#) first, so that you understand how mine is structured (and how it is similar and different to yours).

Then watch the [hardware interface](#) tutorial which shows how to connect that to ros2_control. You should be able to pretty much take my hardware interface and make some small modifications to the class that handles the comms.



JoshNewans

[▶ Jeffrey_Jene](#) Jun '23

Bizarre! Yeah it does sound like a PID controller behaving badly.

One possible issue if you are using my exact code is that the PID settings might have changed since the earlier version? The latest hardware interface is [here](#) and it gives you the ability to tweak the PID parameters.

That version SHOULD leave them as default if you don't set them, but I think there were other versions with a hardcoded value so you may have that?

My other suggestion would be to add print statements into the hardware interface to confirm exactly what is being sent to the motors. That's a little tedious but should get to the bottom of it.



Jeffrey_Jene

Jun '23

It looks like that branch is for Humble. I'm using Foxy, can it be used with it or do I need to upgrade?

[1 reply](#)



Adi

Jun '23

Hey [@JoshNewans](#) thank you for the great tutorial, it has been of great benefit to us as students. We are having difficulty loading the /controller_manager services, we've tried the solutions you suggested earlier in the thread to [@Sasffm](#), but it is still not working. We have the same setup, and the device port has been set correctly, even the user has been added to the dialout group. The following is the error obtained on running the launch file.

```
[INFO] [launch]: All log files can be found below /home/vahin/.ros/log/2023-06-07-09-52-16-628234-ubuntu-13541
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [13544]
[robot_state_publisher-1] Parsing robot urdf xml string.
[robot_state_publisher-1] Link chassis had 2 children
[robot_state_publisher-1] Link caster_wheel had 0 children
[robot_state_publisher-1] Link laser_frame had 0 children
[robot_state_publisher-1] Link left_wheel had 0 children
[robot_state_publisher-1] Link right_wheel had 0 children
[robot_state_publisher-1] [INFO] [1686131537.466259297] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1686131537.466644027] [robot_state_publisher]: got segment caster_wheel
[robot_state_publisher-1] [INFO] [1686131537.466723784] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1686131537.466753468] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1686131537.466781004] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1686131537.466809170] [robot_state_publisher]: got segment right_wheel
[INFO] [ros2_control_node-2]: process started with pid [13606]
[INFO] [spawner.py-3]: process started with pid [13608]
[INFO] [spawner.py-4]: process started with pid [13610]
[ros2_control_node-2] [INFO] [1686131543.680967466] [DiffDriveArduino]: Configuring...
[ros2_control_node-2] [INFO] [1686131543.687668416] [DiffDriveArduino]: Finished Configuration
[ros2_control_node-2] [INFO] [1686131543.688505651] [DiffDriveArduino]: Starting Controller...
[ros2_control_node-2] [INFO] [1686131545.719892550] [controller_manager]: update rate is 30 Hz
[ros2_control_node-2] [INFO] [1686131545.762545973] [controller_manager]: Loading controller 'diff_cont'
[ros2_control_node-2] [ERROR] [1686131545.762714005] [controller_manager]: Loader for controller 'diff_cont' not found
```

```
return.  
[ros2_control_node-2] [INFO] [1686131545.762758207] [controller_manager]: Available classes:  
[ros2_control_node-2] [INFO] [1686131545.762797873] [controller_manager]:   controller_manager/test_controller  
[ros2_control_node-2] [INFO] [1686131545.762828928] [controller_manager]:  
controller_manager/test_controller_failed_init  
[ros2_control_node-2] [INFO] [1686131545.762866667] [controller_manager]:  
controller_manager/test_controller_with_interfaces  
[ros2_control_node-2] [INFO] [1686131545.835433835] [controller_manager]: Loading controller 'joint_broad'  
[ros2_control_node-2] [ERROR] [1686131545.835685402] [controller_manager]: Loader for controller 'joint_broad' not  
found.  
[ros2_control_node-2] [INFO] [1686131545.835744493] [controller_manager]: Available classes:  
[ros2_control_node-2] [INFO] [1686131545.835786603] [controller_manager]:   controller_manager/test_controller  
[ros2_control_node-2] [INFO] [1686131545.835817731] [controller_manager]:  
controller_manager/test_controller_failed_init  
[ros2_control_node-2] [INFO] [1686131545.835846286] [controller_manager]:  
controller_manager/test_controller_with_interfaces  
[ERROR] [spawner.py-4]: process has died [pid 13610, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py  
diff_cont --ros-args'].  
[ERROR] [spawner.py-3]: process has died [pid 13608, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py  
joint_broad --ros-args'].
```

Is there something we've missed out on? The controller_manager had issues earlier as well when running on the dev machine but it was sorted when it was rerun a few times.

[1 reply](#)



JoshNewans

[▶ Adi](#) Jun '23

Ok so the main error here is that it goes to load diff_cont and joint_broad but can't find them. Instead it is finding one called test_controller (I'm not sure if you've put that in or if it's standard?)

Do you have a controllers.yaml file that you are passing in? What is inside it? You should have the diff drive controller and joint state broadcaster defined in there.

If that is set up correctly, then make sure it is actually being loaded!

[1 reply](#)



JoshNewans

[▶ Jeffrey_Jene](#) Jun '23

Yeah the old branch is foxy but I'm not really maintaining it.
The principles are all pretty similar so if you compare the two and watch [this video](#) you should be able to figure it out.

[1 reply](#)



Adi

[▶ JoshNewans](#) Jun '23

Thank you for your prompt response and suggestions, we will look into it.



FrankfurtDynamics

Jun '23

I have a really weird Issue. The robot drives forwards and backwards fine but it doesn't turn.

[1 reply](#)



JoshNewans

[▶ FrankfurtDynamics](#) Jun '23

Hmm that is odd.

Is this with teleoperation or autonomous control?

I'd first use ros2 topic echo to check the contents of your cmd_vel topic are correct and you are sending a Z rotation.

If that is all ok then check your diff drive parameters (wheel spacing etc) are all correct and you don't have anything odd in there like rotational limits.



FrankfurtDynamics

Jun '23

Both it is even a problem when I just use the ROS-Arduino-Bridge.ino and miniterm. When I type 0 255 255 or negative it works fine. When I type 0 255 -255 the motors don't turn at all.

[1 reply](#)



Jeffrey_Jene

[▶ JoshNewans](#) Jun '23

After fooling around with it quite a bit, I do see that it is the PID values, possible something else as well. I could sit and fool with the values, but the behavior is still strange (the motors start slow and rev up to the required speed, but overshoot it and reverse directions). Open loop control works great. I would love to rewrite this thing at some point. I'm fuzzy on a few things I need to research first, like how to translate input tick count to speed to PWM. I'm also not sure what "Ko" is in the PID code. Josh, what are the RPM speeds of your motors? I think mine are about 250 or so, maybe too much for this application (if that even makes a difference). Anyway, I'll keep fussing with it.

[2 replies](#)



Cliff_Andrew_Oliver

Jun '23

```
[INFO] [launch]: All log files can be found below /home/cliff/.ros/log/2023-06-11-09-19-26-129315-cliff-Legion-4342  
[INFO] [launch]: Default logging verbosity is set to INFO  
Task exception was never retrieved  
future: <Task finished name='Task-2' coro=<LaunchService.process_one_event() done, defined at  
/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py:226> exception=PackageNotFoundError("package  
'twist_mux' not found, searching: ['/home/cliff/dev_ws/install/serial_motor_demo',  
'/home/cliff/dev_ws/install/serial_motor_demo_msgs', '/home/cliff/dev_ws/install/diffdrive_arduino',  
'/home/cliff/dev_ws/install/serial', '/home/cliff/dev_ws/install/articubot_one', '/opt/ros/foxy']")>  
Traceback (most recent call last):  
  File "/opt/ros/foxy/lib/python3.8/site-packages/ament_index_python/packages.py", line 50, in get_package_prefix  
    content, package_prefix = get_resource('packages', package_name)  
  File "/opt/ros/foxy/lib/python3.8/site-packages/ament_index_python/resources.py", line 48, in get_resource  
    raise LookupError(  
LookupError: Could not find the resource 'twist_mux' of type 'packages'
```

During handling of the above exception, another exception occurred:

```
Traceback (most recent call last):  
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 228, in _process_one_event  
    await self._process_event(next_event)  
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 248, in __process_event  
    visit_all_entities_and_collect_futures(entity, self.__context))  
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",  
line 45, in visit_all_entities_and_collect_futures  
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)  
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",  
line 45, in visit_all_entities_and_collect_futures  
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)  
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",  
line 45, in visit_all_entities_and_collect_futures
```

```

line 43, in visit_all_entities_and_collect_futures
    futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)
[Previous line repeated 1 more time]
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 38, in visit_all_entities_and_collect_futures
    sub_entities = entity.visit(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/action.py", line 108, in visit
    return self.execute(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/actions/node.py", line 453, in execute
    ret = super().execute(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/actions/execute_process.py", line 823, in execute
    self._expand_substitutions(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/actions/execute_process.py", line 668, in
__expand_substitutions
    cmd = [perform_substitutions(context, x) for x in self.__cmd]
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/actions/execute_process.py", line 668, in <listcomp>
    cmd = [perform_substitutions(context, x) for x in self.__cmd]
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/perform_substitutions_impl.py", line 26, in
perform_substitutions
    return ''.join([context.perform_substitution(sub) for sub in subs])
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/perform_substitutions_impl.py", line 26, in
<listcomp>
    return ''.join([context.perform_substitution(sub) for sub in subs])
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_context.py", line 232, in perform_substitution
    return substitution.perform(self)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/substitutions/executable_in_package.py", line 76, in
perform
    package_prefix = super().perform(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/substitutions/find_package.py", line 79, in perform
    result = self.find(package)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/substitutions/find_package.py", line 96, in find
    return get_package_prefix(package_name)
File "/opt/ros/foxy/lib/python3.8/site-packages/ament_index_python/packages.py", line 52, in get_package_prefix
    raise PackageNotFoundError(
ament_index_python.packages.PackageNotFoundError: "package 'twist_mux' not found, searching:
['/home/cliff/dev_ws/install/serial_motor_demo', '/home/cliff/dev_ws/install/serial_motor_demo_msgs',
'/home/cliff/dev_ws/install/diffdrive_arduino', '/home/cliff/dev_ws/install/serial',
'/home/cliff/dev_ws/install/articubot_one', '/opt/ros/foxy']"
Task exception was never retrieved
future: <Task finished name='Task-8' coro=<LaunchService._process_one_event() done, defined at
/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py:226> exception=RuntimeError('Signal event received
before subprocess transport available.')>
Traceback (most recent call last):
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 228, in _process_one_event
    await self._process_event(next_event)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 248, in __process_event
    visit_all_entities_and_collect_futures(entity, self.__context))
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py",
line 38, in visit_all_entities_and_collect_futures
    sub_entities = entity.visit(context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/action.py", line 108, in visit
    return self.execute(context)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/actions/opaque_function.py", line 75, in execute
    return self._function(context, *self._args, **self._kwargs)
  File "/opt/ros/foxy/lib/python3.8/site-packages/launch/actions/execute_process.py", line 443, in
__on_signal_process_event
    raise RuntimeError('Signal event received before subprocess transport available.')
RuntimeError: Signal event received before subprocess transport available.
[INFO] [robot_state_publisher-1]: process started with pid [4345]
[robot_state_publisher-1] Parsing robot urdf xml string.
[robot_state_publisher-1] Link base_footprint had 0 children
[robot_state_publisher-1] Link chassis had 4 children
[robot_state_publisher-1] Link camera_link had 1 children
[robot_state_publisher-1] Link camera_link_optical had 0 children
[robot_state_publisher-1] Link caster_wheel had 0 children
[robot_state_publisher-1] Link face_link had 0 children
[robot_state_publisher-1] Link laser_frame had 0 children
[robot_state_publisher-1] Link left_wheel had 0 children
[robot_state_publisher-1] Link right_wheel had 0 children
[robot_state_publisher-1] [INFO] [1686455366.472416353] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1686455366.472564150] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1686455366.472588526] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-1] [INFO] [1686455366.472606757] [robot_state_publisher]: got segment camera_link_optical
[robot_state_publisher-1] [INFO] [1686455366.472624637] [robot_state_publisher]: got segment caster_wheel
[robot_state_publisher-1] [INFO] [1686455366.472642309] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1686455366.472659212] [robot_state_publisher]: got segment face_link
[robot_state_publisher-1] [INFO] [1686455366.472675486] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1686455366.472692599] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1686455366.472709362] [robot_state_publisher]: got segment right_wheel
[ERROR] [robot_state_publisher-1]: process[robot_state_publisher-1] failed to terminate '5' seconds after receiving
'SIGINT', escalating to 'SIGTERM'
[INFO] [robot_state_publisher-1]: sending signal 'SIGTERM' to process[robot_state_publisher-1]
[ERROR] [robot_state_publisher-1]: process has died [pid 4345, exit code -15, cmd
'/opt/ros/foxy/lib/robot_state_publisher/robot_state_publisher --ros-args --params-file /tmp/launch_params_zw350_hn'].

```

Cant find whats actually going wrong, feels like its ros not building properly

1 reply



JoshNewans

► FrankfurtDynamics Jun '23

Very odd, I'm suspicious of maybe a wiring issue with the motors/driver/arduino?

Do

- o 255 0
- o -255 0
- o 0 255
- o 0 -255
- All work as expected?

1 reply



JoshNewans

Jun '23

Interesting, I have had similar problems with other motors. I still got them to work but they were a bit more sensitive to the PID settings.



Jeffrey_Jene:

I would love to rewrite this thing at some point

Yeah me too. I didn't write the Arduino code (just made some small mods) and would like to do it from scratch.

-

	Jeffrey_Jene:		
	I'm also not sure what "Ko" is in the PID cod		
	I also found this confusing. I believe this and also the reason everything is done in "encoder ticks per loop" is to avoid doing floating point arithmetic in the Arduino. So the Ko just scales things up. E.g. if we want $K_p = 0.5$ (and $K_o = 1$) we can't do that, so instead we set $K_p = 5$ and $K_o = 10$. It's a little while since I dug into it though so that might be not quite right.		
	Jeffrey_Jene:		
	What are the RPM speeds of your motors? I think mine are about 250 or so, maybe too much for this application (if that even makes a difference).		
	Pretty sure they are 110RPM. 250 is pretty fast, especially if your encoders aren't super fast. E.g. your requests might boil down to the difference between a few integers in counts per loop and the PID will be jumping rapidly between them.		
	2 replies		
	JoshNewans	▶ Cliff_Andrew_Oliver	Jun '23
	Make sure when posting code blocks (I have edited yours) you put backticks (the ~ key without shift) around it to make it easier to read, like this: <pre>... My_code_here ...</pre> Your issue here is you don't have <code>twist_mux</code> installed (<code>sudo apt install ros-foxy-twist-mux</code>). This error should be clearer, except I have not explicitly made <code>twist_mux</code> a dependency of <code>articubot_one</code> so it tries to run without checking that it exists, and you get that very messy error.		
	FrankfurtDynamics	▶ JoshNewans	Jun '23
	I found the problem it seems to be some internal problem with the Arduino uno, it works fine on the nano.		
	Jeffrey_Jene	▶ JoshNewans	Jun '23
	I think I will take a stab at writing new Arduino code. Josh, if you were personally to do this would you continue using Serial or would you go for something like MicroROS (keeping in mind one could no longer use the Nano if that were the case)? I use the Teensy quite a bit so perhaps I'll try something for both (I'd like to learn MicroROS myself). This is out of scope so I'll start a new topic for discussion later.		
	Jeffrey_Jene	▶ JoshNewans	Jun '23
	Hey Josh, wanted to let you know I purchased a couple of 100RPM motors and that seemed to do the trick. Everything is now running smoothly. Those other motors were probably just too fast for the PID/encoders to handle. BTW, is it possible to get permissions to create new topics? It doesn't look like I have (unless I'm not supposed to).		
	1 reply		
	JoshNewans	▶ Jeffrey_Jene	Jun '23
	Nice, I'm very glad that sorted it (it's always disappointing when you sink money into extra hardware and it doesn't help!) Yeah, creating new topics is restricted to paid members over at Patreon however I have tried to make the bottom tier very affordable. This also gets you access to the Discord server and helps support the channel. I haven't got Patreon and Discourse synced up yet though and have to update user permissions manually, so don't be afraid to remind me if needed 😊		
	1 reply		
	Jeffrey_Jene	▶ JoshNewans	Jun '23
	Sounds fair. 😊 I have now joined Patreon.		
	Jeffrey_Jene		Jul '23
	Maybe the answer to this is in my face and I don't see it, but I'm going to ask anyway. I've created a launch file for the physical robot after success with all the simulations. It runs and teleoperates just fine. When I open RVIZ, however, I noticed that the model's wheels are screwed up because the wheel transforms can't be resolved (we used to cheat and run <code>joint_state_publisher_gui</code> to resolve this). This does not happen when running RVIZ with the Gazebo simulation. Is this behavior correct? Is there something I can do (besides running the JSP gui) to fix this issue? Edit: I figured it out. I had the wheels parented to the <code>base_link</code>. When I changed the parent to the chassis the joints are now recognized. I'm leaving this post for anyone who might have the same issue with their design.		
	1 reply		
	Devbot		Jul '23
	Im having issues where my Twist msgs from teleop is having the following problems: -not updating my robot position correctly -unstable, moves slow sometimes and very fast other times -when moving straight, it never updates the position except for a fixed 0.1 meter even if i keep going straight for more. -after a few turns, it gets stuck and i cant use teleop anymore until i manually nudge the wheel. note: the updates and revolutions are exact when im turning the wheels using my hands. the problems start when i use teleop. and i am also confused about where to change my PID values because there are ones in the <code>diffdrive_arduino.cpp</code> (<code>arduino_.setPidValues(30,20,0,100)</code>) and ones in the arduino file, which ones should i be changing? All the code works up until the gazebo simulation and mapping. im using ros humble and have updates my files appropriately using your most recent video. Any help as to where to start fixing the issue would be very very appreciatedddd FIXED!! = The issue was very very lame, even though i spent days changing the code assuming its the cause. The issue was wiring and power! the encoders and logic were getting the power from the controller because i forgot to provide it power and ground it from and to the arduino, and it caused it to keep shutting off and on and acting crazy.		
	1 reply		
	JoshNewans	▶ Jeffrey_Jene	Jul '23
	Glad you got it sorted. I was going to suggest checking that you were running the <code>joint_state_broadcaster</code> which is responsible for publishing the joint states when using <code>ros2_control</code>, which are then read by <code>robot_state_publisher</code> to broadcast the transforms. Including that as a note in case someone has a similar (but different!) issue to what you did 😊		
	JoshNewans	▶ Devbot	Jul '23
	Ahh yep those kiind of wiring issues can be a real pain and cause very sporadic behaviour (or kill your encoders as some people have found 😞) I'm glad it's fixed for you though. By the PID values from memory the <code>FrankfurtDynamics</code> should change the <code>base_drive</code> fixture to the <code>base</code> and <code>base_drive</code> to the <code>base</code> and <code>base_drive</code> to the <code>base</code>.		

re the PID values, from memory on the Foxy version you should change the hardcoded values in the cpp, and on Humble you can set them as a parameter. If you change them on the Arduino I think that will be changing the default value which is the case if diffdrive_arduino never sends an override.

(I think that's all correct, I haven't looked at it for a little while)



Linh_Nguy_n_D_ng

Aug '23

Hi @JoshNewans

I installed RO2 Humble. I am getting this error when trying to build diffdrive_arduino with colcon build. Can you help me?

```
Starting >>> serial
Starting >>> articubot_one
Finished <<< articubot_one [0.88s]
Finished <<< serial [1.39s]
Starting >>> diffdrive_arduino
— stderr: diffdrive_arduino
In file included from /home/robot/rasp_ws/src/diffdrive_arduino/src/fake_robot.cpp:1:
/home/robot/rasp_ws/src/diffdrive_arduino/include/diffdrive_arduino/fake_robot.h:7:10:
fatal error: hardware_interface/base_interface.hpp: No such file or directory
7 | #include "hardware_interface/base_interface.hpp"
  | ^~~~~~
compilation terminated.
gmake[2]: *** [CMakeFiles/fake_robot.dir/build.make:76:
CMakeFiles/fake_robot.dir/src/fake_robot.cpp.o] Error 1
gmake[1]: *** [CMakeFiles/Makefile2:165: CMakeFiles/fake_robot.dir/all] Error 2
gmake[1]: *** Waiting for unfinished jobs...
In file included from
/home/robot/rasp_ws/src/diffdrive_arduino/src/diffdrive_arduino.cpp:1:
/home/robot/rasp_ws/src/diffdrive_arduino/include/diffdrive_arduino/diffdrive_arduino.h:7:
10: fatal error: hardware_interface/base_interface.hpp: No such file or directory
7 | #include "hardware_interface/base_interface.hpp"
  | ^~~~~~
compilation terminated.
gmake[2]: *** [CMakeFiles/diffdrive_arduino.dir/build.make:76:
CMakeFiles/diffdrive_arduino.dir/src/diffdrive_arduino.cpp.o] Error 1
gmake[2]: *** Waiting for unfinished jobs...
gmake[1]: *** [CMakeFiles/Makefile2:139: CMakeFiles/diffdrive_arduino.dir/all] Error 2
gmake: *** [Makefile:146: all] Error 2
```

Failed <<< diffdrive_arduino [23.8s, exited with code 2]

Summary: 2 packages finished [26.4s]

1 package failed: diffdrive_arduino

1 package had stderr output: diffdrive_arduino

[1 reply](#)



Linh_Nguy_n_D_ng

[▶ Linh_Nguy_n_D_ng](#) Aug '23

@JoshNewans . I installed diffdrive_arduino of [GitHub - Buzzology/diffdrive_arduino: A simple ROS Control Hardware Interface for differential drive control of robot whose motors are connected via an Arduino \(or similar microcontroller\)](#).. I can run teleop_twist_keyboard to drive my around. My motors can work. But when I open rviz2 for tracking, when I press some letters 'i' and 'm' on the keyboard, the robot runs circle around base_link , doesn't go straight (forward and backward). I installed ROS2 HUMBLE. Can you help me, plz?



Shai_Mizrachi

Sep '23

I'm a bit confuse

When using SIM mode

The file ros2_control.xacro define the differential wheel plug in for gazebo through the yaml files

the file gazebo_control.xacro define diff_drive plugin

isn't redundant? what do I miss with my understanding?



mamba0206

Sep '23

@JoshNewans

Hi! Firstly, amazing tutorial which clearly explains how to interface to a real robot. However, I am facing an issue when launching the real robot. The error is as below

```
[ros2_control_node-2] [INFO] [1694504172.944274701] [resource_manager]: Loading hardware 'RealRobot'
[ros2_control_node-2] [INFO] [1694504172.946497851] [resource_manager]: Initialize hardware 'RealRobot'
[ros2_control_node-2] terminate called after throwing an instance of 'std::invalid_argument'
```

I figured the error occurs in the ros2_control/hardware_interface/resource_manager.cpp file. The specific line where the error occurs I think is this:
const rcpp_lifecycle::State new_state = hardware.initialize(hardware_info);

I'm using an Arduino Mega instead. I have tested my bot with the ros_arduino_bridge package and it works fine when running miniterm. Is there a way to resolve this? Thank you!



robotmaker

Sep '23

Hi, thank you for all your work on those videos, which I love learning from.

I'm trying to create my own robot under ROS 2 Humble. It seems that I have successfully managed to use Nav2. However, I don't understand how ROS 2 Control is used by Nav2. What does Nav2 Controller Server and Planner use to work? And finally, the most important question: Is there any controller from ROS 2 Control for controlling a 3-wheel omnidirectional robot?



tadashi

Sep '23

Hey Josh,

First of all, thanks for all these great tutorials! They are helping me loads during my studies and thanks for replying. This is for a school project, so I appreciate all the help I can get.

I get the same error message as Samet_Elmaci in this thread:

[Discussion - Changes for running mobile robot on Humble/Jammy - #14 by Samet_Elmaci](#) .

Some information about my setup:

I am using ros2 humble in a docker container. I have the exact same hardware setup as you for ease of following your tutorials. I have forwarded the USB ports on the pi to the container, so I am able to get lidar data and control the motors. Therefore the container setup itself is not likely the

issue.

I have cloned the repo for articubot_one (humble branch) and the error happens when i try to run the the launch_robot.launch.py. I have tried various things to fix the issue:

- I can control the motors with miniterm.
- I have your diffdrive_arduino pkg (humble branch).
- I have your serial pkg (newans_ros2 branch).
- I have tried running it both on my dev machine and the rpi.
- Following packages are installed:
 - ros-humble-ros2-controllers
 - ros-humble-ros2-control
 - ros-humble-gazebo-ros2-control (on dev machine)
- ros2 control works when running 'launch_sim.launch.py'. (I can teleop).
- I have of course rebuild with colcon and sourced the workspace.
- I tried referencing the serial device in the ros2_controller.xacro file both by /dev/ttyUSB0 and with /dev/serial/by-id/{the id i found for the nano}.
- I have changed to both read, write, execute privilege for the serial port.
- User is added to the dialout group
- I have tried something as simple as sudo apt-get update and upgrade.

Things i am going to try:

- I have only wired one of the motors up, so i will connect the other one as well.
- Try foxy instead of humble.
- Install ros directly on the machines instead of docker containers.

Error message:

```
devuser@pi:overlay_ws$ ros2 launch articubot_one launch_robot.launch.py
[INFO] [launch]: All log files can be found below /home/devuser/.ros/log/2023-09-26-11-54-50-619436-pi-2661
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [2663]
[INFO] [twist_mux-2]: process started with pid [2665]
[twist_mux-2] [INFO] [1695729293.661287767] [twist_mux]: Topic handler 'topics.joystick' subscribed to topic 'cmd_vel_joy': timeout = 0.500000s , priority = 100.
[twist_mux-2] [INFO] [1695729293.669544297] [twist_mux]: Topic handler 'topics.navigation' subscribed to topic 'cmd_vel': timeout = 0.500000s , priority = 10.
[twist_mux-2] [INFO] [1695729293.671005906] [twist_mux]: Topic handler 'topics.tracker' subscribed to topic 'cmd_vel_tracker': timeout = 0.500000s , priority = 20.
[robot_state_publisher-1] [INFO] [1695729293.707919896] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1695729293.708507536] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1695729293.708640478] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-1] [INFO] [1695729293.708740736] [robot_state_publisher]: got segment camera_link_optical
[robot_state_publisher-1] [INFO] [1695729293.708833883] [robot_state_publisher]: got segment caster_wheel
[robot_state_publisher-1] [INFO] [1695729293.708923104] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1695729293.709009436] [robot_state_publisher]: got segment face_link
[robot_state_publisher-1] [INFO] [1695729293.709094694] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1695729293.709181637] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1695729293.709267247] [robot_state_publisher]: got segment right_wheel
[INFO] [ros2_control_node-3]: process started with pid [2700]
[INFO] [spawner-4]: process started with pid [2702]
[INFO] [spawner-5]: process started with pid [2704]
[ros2_control_node-3] [INFO] [1695729302.263149623] [resource_manager]: Loading hardware 'RealRobot'
[ros2_control_node-3] terminate called after throwing an instance of 'pluginlib::LibraryLoadException'
[ros2_control_node-3] what(): According to the loaded plugin descriptions the class diffdrive_arduino/DiffDriveArduino with base class type hardware_interface::SystemInterface does not exist. Declared types are diffdrive_arduino/DiffDriveArduinoHardware
fake_components/GenericSystem mock_components/GenericSystem testHardware_components/TestSystemCommandModes
testHardware_components/TestTwoJointSystem test_system
[ros2_control_node-3] Stack trace (most recent call last):
[ros2_control_node-3] #16 Object "/usr/lib/aarch64-linux-gnu/ld-linux-aarch64.so.1", at 0xffffffffffffff, in
[ros2_control_node-3] #15 Object "/opt/ros/humble/lib/controller_manager/ros2_control_node", at 0xaaaae2d52c6f, in
[ros2_control_node-3] #14 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffffa63274cb, in __libc_start_main
[ros2_control_node-3] #13 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffffa63273fb, in
[ros2_control_node-3] #12 Object "/opt/ros/humble/lib/controller_manager/ros2_control_node", at 0xaaaae2d52767, in
[ros2_control_node-3] #11 Object "/opt/ros/humble/lib/libcontroller_manager.so", at 0xffffa699cba7, in
controller_manager::ControllerManager::ControllerManager(std::shared_ptr<rcpp::Executor, std::cxx11::basic_string<char, std::char_traits, std::allocator > const&, std::cxx11::basic_string<char, std::char_traits, std::allocator > const&, rclcpp::NodeOptions const&)
[ros2_control_node-3] #10 Object "/opt/ros/humble/lib/libcontroller_manager.so", at 0xffffa699bfcf, in
controller_manager::ControllerManager::init_resource_manager(std::cxx11::basic_string<char, std::char_traits, std::allocator > const&)
[ros2_control_node-3] #9 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0xffffa6197237, in
hardware_interface::ResourceManager::load_urdf(std::cxx11::basic_string<char, std::char_traits, std::allocator > const&, bool)
[ros2_control_node-3] #8 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0xffffa61959e3, in
[ros2_control_node-3] #7 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0xffffa61a16b7, in
[ros2_control_node-3] #6 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffffa6582e03, in __cxa_throw
[ros2_control_node-3] #5 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffffa6582b1f, in std::terminate()
[ros2_control_node-3] #4 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffffa6582abb, in
[ros2_control_node-3] #3 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffffa65862db, in __gnu_cxx::__verbose_terminate_handler()
[ros2_control_node-3] #2 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffffa632712f, in abort
[ros2_control_node-3] #1 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffffa633a67b, in raise
[ros2_control_node-3] #0 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffffa637f200, in
[ros2_control_node-3] Aborted (Signal sent by kill()) 2700 1000
[ERROR] [ros2_control_node-3]: process has died [pid 2700, exit code -6, cmd '/opt/ros/humble/lib/controller_manager/ros2_control_node --ros-args --params-file /tmp/launch_params_navrtmcw --params-file /overlay_ws/install/articubot_one/share/articubot_one/config/my_controllers.yaml'].
[spawner-4] [INFO] [1695729307.338964270] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-5] [INFO] [1695729307.421576180] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
[spawner-4] [INFO] [1695729309.405147189] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-5] [INFO] [1695729309.489268521] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
[spawner-4] [INFO] [1695729311.472865011] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-5] [INFO] [1695729311.556779495] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
[spawner-4] [INFO] [1695729313.543320664] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-5] [INFO] [1695729313.625254842] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
[spawner-4] [ERROR] [1695729315.200286903] [spawner_joint_broad]: Controller manager not available
[spawner-5] [ERROR] [1695729315.279275328] [spawner_diff_cont]: Controller manager not available
[ERROR] [spawner-4]: process has died [pid 2702, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner joint_broad --ros-args'].
[ERROR] [spawner-5]: process has died [pid 2704, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner diff_cont --ros-args'].
```



Mikhail Andryievich

Oct '23

Hi,
I'm running directly on arduino and have the same issue.

```
ros2@robot:/home/robot_ws$ ros2 launch lunorobot launch_robot.launch.py
[INFO] [launch]: All log files can be found below /home/robot2/.ros/log/2023-10-03-17-12-02-039075-robot-4349
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [4350]
[robot_state_publisher-1] [INFO] [1696345922.974149735] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1696345922.974446916] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1696345922.974494767] [robot_state_publisher]: got segment caster_wheel
[robot_state_publisher-1] [INFO] [1696345922.974526341] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1696345922.974552155] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1696345922.974576322] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1696345922.974600988] [robot_state_publisher]: got segment right_wheel
[INFO] [ros2_control_node-2]: process started with pid [4388]
[INFO] [spawner-3]: process started with pid [4390]
[INFO] [spawner-4]: process started with pid [4392]
[ros2_control_node-2] [INFO] [1696345928.903734258] [resource_manager]: Loading hardware 'RealRobot'
[ros2_control_node-2] terminate called after throwing an instance of 'pluginlib::LibraryLoadException'
[ros2_control_node-2] what(): According to the loaded plugin descriptions the class diffdrive_arduino/DiffDriveArduino with base class type
```

```
[ros2_control_node-2] Warning: According to the ros2_control plugin descriptions the class interface::SystemInterface does not exist. Declared types are diffdrive_arduino/DiffDriveArduinoHardware
fake_components/GenericSystem mock_components/GenericSystem test_hardware_components/TestSystemCommandModes
test_hardware_components/TestTwoJointSystem test_system
[ros2_control_node-2] Stack trace (most recent call last):
[ros2_control_node-2] #16 Object "/usr/lib/aarch64-linux-gnu/ld-linux-aarch64.so.1", at 0xffffffffffff, in
[ros2_control_node-2] #15 Object "/opt/ros/humble/lib/controller_manager/ros2_control_node", at 0xaaaaaa4c2c6f, in
[ros2_control_node-2] #14 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffff9fb674cb, in __libc_start_main
[ros2_control_node-2] #13 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffff9fb673fb, in
[ros2_control_node-2] #12 Object "/opt/ros/humble/lib/controller_manager/ros2_control_node", at 0xaaaaaa4c2767, in
[ros2_control_node-2] #11 Object "/opt/ros/humble/lib/libcontroller_manager.so", at 0xffffa01dd717, in
controller_manager::ControllerManager::ControllerManager(std::shared_ptr<rcpp::Executor, std::cxx11::basic_string<char, std::char_traits,
std::allocator> const&, std::cxx11::basic_string<char, std::char_traits, std::allocator> const&, rcpp::NodeOptions const&)
[ros2_control_node-2] #10 Object "/opt/ros/humble/lib/libcontroller_manager.so", at 0xffffa01dcb3f, in
controller_manager::ControllerManager::init_resource_manager(std::cxx11::basic_string<char, std::char_traits, std::allocator> const&)
[ros2_control_node-2] #9 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0xffff9fd7237, in
hardware_interface::ResourceManager::load_urdf(std::cxx11::basic_string<char, std::char_traits, std::allocator> const&, bool)
[ros2_control_node-2] #8 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0xffff9fd59e3, in
[ros2_control_node-2] #7 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0xffff9fd59e16b7, in
[ros2_control_node-2] #6 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffff9fdc2e03, in __cxa_throw
[ros2_control_node-2] #5 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffff9fdc2b1f, in std::terminate()
[ros2_control_node-2] #4 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffff9fdc2abb, in
[ros2_control_node-2] #3 Object "/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.30", at 0xffff9fdc62db, in __gnu_cxx::__verbose_terminate_handler()
[ros2_control_node-2] #2 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffff9fb6712f, in abort
[ros2_control_node-2] #1 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffff9fb7a67b, in raise
[ros2_control_node-2] #0 Object "/usr/lib/aarch64-linux-gnu/libc.so.6", at 0xffff9fbf200, in
[ros2_control_node-2] Aborted (Signal sent by tkill() 4388 1000)
[ERROR] [ros2_control_node-2]: process has died [pid 4388, exit code -6, cmd '/opt/ros/humble/lib/controller_manager/ros2_control_node --ros-
args --params-file /tmp/launch_params_izmqtko --params-file /home/robot_ws/install/unorobot/share/unorobot/config/my_controllers.yaml'].
[spawner-3] [INFO] [1696345932.213156760] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-4] [INFO] [1696345932.541984436] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
[spawner-3] [INFO] [1696345934.271893949] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-4] [INFO] [1696345934.602429954] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
[spawner-3] [INFO] [1696345936.328438815] [spawner_joint_broad]: Waiting for 'controller_manager' node to exist
[spawner-4] [INFO] [1696345936.660799370] [spawner_diff_cont]: Waiting for 'controller_manager' node to exist
```

Could you please help?

1 reply



Mikhail Andryievich

Oct '23

Mikhail Andryievich:
with base class type hardware_interface::SystemInterface does not exist.

Hi,

Solution found for Humble version
There are 3 issues in ros2_control.xacro file:

1. the name of the plugin should be diffdrive_arduino/DiffDriveArduinoHardware
2. the name of the timeout parameter should be timeout_ms
3. the sequence of the state_interface should be position then velocity, see correct xml code below:

```
<ros2_control name="RealRobot" type="system">
  <hardware>
    <plugin>diffdrive_arduino/DiffDriveArduinoHardware</plugin>
    <param name="left_wheel_name">left_wheel_joint</param>
    <param name="right_wheel_name">right_wheel_joint</param>
    <param name="loop_rate">30</param>
    <param name="device">/dev/ttyUSB0</param>
    <param name="baud_rate">57600</param>
    <param name="timeout_ms">1000</param>
    <param name="enc_counts_per_rev">17280</param>
  </hardware>
  <!-- Note everything below here is the same as the Gazebo one -->
  <joint name="left_wheel_joint">
    <command_interface name="velocity">
      <param name="min">-10</param>
      <param name="max">10</param>
    </command_interface>
    <state_interface name="position"/>
    <state_interface name="velocity"/>
  </joint>
  <joint name="right_wheel_joint">
    <command_interface name="velocity">
      <param name="min">-10</param>
      <param name="max">10</param>
    </command_interface>
    <state_interface name="position"/>
    <state_interface name="velocity"/>
  </joint>
</ros2_control>
</robot>
```

1 reply



LeoCleaner

Oct '23

Hi Josh,
I find your videos awesome!!!
Would you mind to plan a topic also using some kind of 9dof sensor (for example bno055) for better navigation? Are they really needed or SLAM is enough?

Thank you very much!

Cheers,
Ivo



The-Kriz

Oct '23

hi im new to Ros and needed some help. From the tutorials i have leard few thing and iwas able to build a 4WD diff_drive robot ([GitHub](#) - [The-Kriz/rog_ros_bot](#))

but was able to run the bot in sim, but didnt have the luck to make the actual bot run as i couidnt figure out how to write code for the hardware part (i only know basic arduino code)

can someone help me to convert or let me know what all i should change from the provided code for 2WD diff_drive to 4WD diff_drive

thanks you,
Kriz



Mano

Oct '23

Hi

...
It was a great lecture, thanks for sharing with us. I was able to follow most part of it, but i was unable to understand how input and output ports are connected in the hardware interface plugin.
thanks a lot for sharing with us.



Robotawi

Hi [@JoshNewans](#)

Thank you for the great work!

I have a question and would appreciate your help.

Could you please explain the rationale behind the specific formula used in the `setMotorValues` method for converting angular velocity to encoder counts per control loop iteration?

```
arduino_.setMotorValues(l_wheel_.cmd / l_wheel_.rads_per_count / cfg_.loop_rate, r_wheel_.cmd / r_wheel_.rads_per_count / cfg_.loop_rate);
```

Specifically, why are the desired angular velocities divided by both the `rads_per_count` conversion factor and the `loop_rate` ?

If I were to adapt this code for a different robot with different hardware or control requirements, what aspects of this line would I need to modify?

Oct '23



eaganrodegard

I seem to be having a problem with the `teleop_twist_joy` node not auto-repeating.

I used `ros2 topic echo /diff_cont/cmd_vel_unstamped` to monitor the command being sent when I was using the joy stick to control the robot.

Initial I would see the command and the robot would move. But after a few seconds, no more commands would be sent even when the joystick is held full forward with the enable button pressed.

I have the `'autorepeat_rate'` set to 20.0 in the `joystick.yaml` file.

What am I missing?

Any ideas would be welcome. Thank you

Nov '23



nam_Duong

I watched the video "You can use ANY hardware with `ros2_control`", I downloaded the code of that video, The purpose of that code is that I want to use the frame available in the code to connect to the arduino mega board. Currently I am in the stage of connecting `ros2_control` to arduino through the code framework available in the video, But I have encountered problems when launching the `diffbot.launch.py` file, currently I am working on the `ros_humble` version. I'm really sorry that my English sentence seems grammatically incorrect. Thank you everyone for your help. I just edited the device parameter:

```
/dev/serial/by-path/pci-0000:05:00.3-usb-0:2:1.0 error: namdc@Nam20195107:~/project_arduino$ ros2 launch diffdrive_arduino diffbot.launch.py
[INFO] [launch]: All log files can be found below /home/namdc/.ros/log/2023-11-25-12-52-15-389354-Nam20195107-10120 [INFO] [launch]: Default logging verbosity is set to INFO [INFO] [ros2_control_node-1]: process started with pid [10123] [INFO] [robot_state_publisher-2]: process started with pid [10125] [INFO] [spawner-3]: process started with pid [10127] [ros2_control_node-1] [INFO] [1700891536.231110072] [resource_manager]: Loading hardware 'DiffBot' [ros2_control_node-1] [INFO] [1700891536.232843117] [resource_manager]: Initialize hardware 'DiffBot' [ros2_control_node-1] [INFO] [1700891536.233266397] [resource_manager]: Successful initialization of hardware 'DiffBot' [ros2_control_node-1] [INFO] [1700891536.233504397] [resource_manager]: 'configure' hardware 'DiffBot' [ros2_control_node-1] [INFO] [1700891536.233545464] [DiffDriveArduinoHardware]: Configuring ...please wait... [ros2_control_node-1] terminate called after throwing an instance of 'LibSerial::OpenFailed' [ros2_control_node-1] what(): Bad file descriptor [ros2_control_node-1] Stack trace (most recent call last): [ros2_control_node-1] [WARN] [1700891536.235810194] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a root link with an inertia. As a workaround, you can add an extra dummy link to your URDF. [robot_state_publisher-2] [INFO] [1700891536.236364170] [robot_state_publisher]: got segment base_link [robot_state_publisher-2] [INFO] [1700891536.236486892] [robot_state_publisher]: got segment caster_frontal_wheel [robot_state_publisher-2] [INFO] [1700891536.236504515] [robot_state_publisher]: got segment caster_rear_wheel [robot_state_publisher-2] [INFO] [1700891536.236517450] [robot_state_publisher]: got segment left_wheel [robot_state_publisher-2] [INFO] [1700891536.236529713] [robot_state_publisher]: got segment right_wheel [ros2_control_node-1] #22 Object "", at 0xffffffffffffff, in [ros2_control_node-1] #21 Object "/opt/ros/humble/lib/controller_manager/ros2_control_node", at 0x55ebb695dd84, in [ros2_control_node-1] #20 Source ".../csu/libc-start.c", line 392, in __libc_start_main_impl [0x7faad2c29e3f] [ros2_control_node-1] #19 Source ".../sysdeps/nptl/libc_start_call_main.h", line 58, in __libc_start_call_main [0x7faad2c29d8f] [ros2_control_node-1] #18 Object "/opt/ros/humble/lib/controller_manager/ros2_control_node", at 0x55ebb695d89e, in [ros2_control_node-1] #17 Object "/opt/ros/humble/lib/libcontroller_manager.so", at 0x7faad3657661, in controller_manager::ControllerManager:ControllerManager(std::shared_ptr, std::cxx11::basic_string<char, std::char_traits, std::allocator> > const&, std::cxx11::basic_string<char, std::char_traits, std::allocator> > const&, rclcpp::NodeOptions const&) [ros2_control_node-1] #16 Object "/opt/ros/humble/lib/libcontroller_manager.so", at 0x7faad3656b2e, in controller_manager::ControllerManager::init_resource_manager(std::cxx11::basic_string<char, std::char_traits, std::allocator> > const&) [ros2_control_node-1] #15 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0x7faad32f8900, in hardware_interface::ResourceManager::activate_all_components() [ros2_control_node-1] #14 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0x7faad32f80bb, in hardware_interface::ResourceManager::set_component_state(std::cxx11::basic_string<char, std::char_traits, std::allocator> > const&, rclcpp_lifecycle::State&) [ros2_control_node-1] #13 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0x7faad32f95e2, in bool hardware_interface::ResourceManager::set_component_state(hardware_interface::System&, rclcpp_lifecycle::State const&) [ros2_control_node-1] #12 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0x7faad330489b, in [ros2_control_node-1] #11 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0x7faad331aaeb, in [ros2_control_node-1] #10 Object "/opt/ros/humble/lib/libhardware_interface.so", at 0x7faad3312776, in hardware_interface::System::configure() [ros2_control_node-1] #9 Object "/home/namdc/project_arduino/install/diffdrive_arduino/lib/libdiffdrive_arduino.so", at 0x7faad614b24, in diffdrive_arduino::DiffDriveArduinoHardware::on_configure(rclcpp_lifecycle::State const&) [ros2_control_node-1] #8 Object "/home/namdc/project_arduino/install/diffdrive_arduino/lib/libdiffdrive_arduino.so", at 0x7faad61765c, in ArduinoComms::connect(std::cxx11::basic_string<char, std::char_traits, std::allocator> > const&, int, int) [ros2_control_node-1] #7 Object "/usr/lib/x86_64-linux-gnu/libserial.so.1.0.0", at 0x7faad5a40b3, in [ros2_control_node-1] #6 Object "/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.30", at 0x7faad30ae4d7, in __cxa_throw [ros2_control_node-1] #5 Object "/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.30", at 0x7faad30ae276, in std::terminate() [ros2_control_node-1] #4 Object "/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.30", at 0x7faad30ae20b, in [ros2_control_node-1] #3 Object "/usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.30", at 0x7faad30a2b9d, in [ros2_control_node-1] #2 Source ".../stdlib/abort.c", line 79, in abort [0x7faad2c287f2] [ros2_control_node-1] #1 Source ".../sysdeps/posix/raise.c", line 26, in raise [0x7faad2c247f5] [ros2_control_node-1] #0 | Source ".../nptl/thread_kill.c", line 89, in __pthread_kill_internal [ros2_control_node-1] | Source ".../nptl/thread_kill.c", line 78, in __pthread_kill_implementation [ros2_control_node-1] Source ".../nptl/thread_kill.c", line 44, in __pthread_kill [0x7faad2c969fc] [ros2_control_node-1] Aborted (Signal sent by tkill() 10123 1000) [ERROR] [ros2_control_node-1]: process has died [pid 10123, exit code -6, cmd '/opt/ros/humble/lib/controller_manager/ros2_control_node --ros-args --params-file /tmp/launch_params_xpux93li --params-file /home/namdc/project_arduino/install/diffdrive_arduino/share/diffdrive_arduino/config/diffbot_controllers.yaml']. [spawner-3] [INFO] [1700891538.812032991] [spawner_joint_state_broadcaster]: Waiting for '/controller_manager' node to exist [spawner-3] [INFO] [1700891540.830612800] [spawner_joint_state_broadcaster]: Waiting for '/controller_manager' node to exist [spawner-3] [INFO] [1700891542.849569356] [spawner_joint_state_broadcaster]: Waiting for '/controller_manager' node to exist [spawner-3] [INFO] [1700891544.867227744] [spawner_joint_state_broadcaster]: Waiting for '/controller_manager' node to exist [spawner-3] [ERROR] [1700891546.883473615] [spawner_joint_state_broadcaster]: Controller manager not available [ERROR] [spawner-3]: process has died [pid 10127, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner joint_state_broadcaster --controller-manager /controller_manager --ros-args']. [INFO] [spawner-4]: process started with pid [10167] [INFO] [rviz2-5]: process started with pid [10169] [rviz2-5] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway. [rviz2-5] [INFO] [1700891547.947906029] [rviz2]: Stereo is NOT SUPPORTED [rviz2-5] [INFO] [1700891547.948230081] [rviz2]: OpenGl version: 4.6 (GLSL 4.6) [rviz2-5] [INFO] [1700891547.987819732] [rviz2]: Stereo is NOT SUPPORTED [rviz2-5] [ERROR] [1700891548.258911760] [rviz2]: The link is static or has unrealistic inertia, so the equivalent inertia box will not be shown. [rviz2-5] [ERROR] [1700891548.259194124] [rviz2]: The link is static or has unrealistic inertia, so the equivalent inertia box will not be shown. [rviz2-5] [spawner-4] [INFO] [1700891549.758581685] [spawner_diffbot_base_controller]: Waiting for '/controller_manager' node to exist [spawner-4] [INFO] [1700891551.777160247] [spawner_diffbot_base_controller]: Waiting for '/controller_manager' node to exist [spawner-4] [INFO] [1700891553.793324662] [spawner_diffbot_base_controller]: Waiting for '/controller_manager' node to exist [spawner-4] [INFO] [1700891555.810854949] [spawner_diffbot_base_controller]: Waiting for '/controller_manager' node to exist [spawner-4] [ERROR] [1700891557.828001767] [spawner_diffbot_base_controller]: Controller manager not available [ERROR] [spawner-4]: process has died [pid 10167, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner diffbot_base_controller --controller-manager /controller_manager --ros-args'].
```

Nov '23



Shah

Thank you very much [@JoshNewans](#) for such a useful content! I'm facing one issue. My real robot wheels are moving fine but in `rviz2` it is not moving. The tf is also not moving. Fixed frame is 'odom'. Tried other fixed frame but of no avail. Don't know the real problem. If anyone knows about this issue please help me out. Thanks once again.

Nov '23

**Mostafa**

Nov '23

Hi @JoshNewans I have learned a lot from your tutorials and applying them on my graduation project so thank you for all the work you have done I would not get anywhere without your generous help
I have been running into issues when I run the "launch_robot.launch.py" on the RaspberryPi while it works perfectly on my laptop and in both cases the Arduino board works fine but I do not seem to load "diff_cont" for some reason on the Pi here is the error I get

```
[INFO] [ros2_control_node-4]: process started with pid [196016]
[INFO] [spawner-5]: process started with pid [196018]
[INFO] [spawner-6]: process started with pid [196020]
[ros2_control_node-4] [INFO] [1700919705.073008620] [resource_manager]: Loading hardware 'RealRobot'
[ros2_control_node-4] [INFO] [1700919705.094977924] [resource_manager]: Initialize hardware 'RealRobot'
[ros2_control_node-4] [INFO] [1700919705.098197162] [DiffDriveArduino]: Configuring...
[ros2_control_node-4] [INFO] [1700919705.128308250] [DiffDriveArduino]: Finished Configuration
[ros2_control_node-4] [INFO] [1700919705.128477832] [resource_manager]: Successful initialization of hardware
'RealRobot'
[ros2_control_node-4] [INFO] [1700919705.129383249] [resource_manager]: 'configure' hardware 'RealRobot'
[ros2_control_node-4] [INFO] [1700919705.129478457] [resource_manager]: Successful 'configure' of hardware 'RealRobot'
[ros2_control_node-4] [INFO] [1700919705.129508218] [resource_manager]: 'activate' hardware 'RealRobot'
[ros2_control_node-4] [INFO] [1700919705.129532016] [DiffDriveArduino]: Starting Controller...
[ros2_control_node-4] [INFO] [1700919706.134547443] [resource_manager]: Successful 'activate' of hardware 'RealRobot'
[ros2_control_node-4] [INFO] [1700919706.271511173] [controller_manager]: update rate is 30 Hz
[ros2_control_node-4] [INFO] [1700919706.28859315] [controller_manager]: RT kernel is recommended for better
performance
[ros2_control_node-4] [INFO] [1700919707.306182869] [controller_manager]: Loading controller 'joint_broad'
[ros2_control_node-4] [INFO] [1700919707.525818714] [controller_manager]: Loading controller 'diff_cont'
[ros2_control_node-4] [ERROR] [1700919707.525999760] [controller_manager]: Loader for controller 'diff_cont' (type
'diff_drive_controller/DiffDriveController') not found.
[ros2_control_node-4] [INFO] [1700919707.526040966] [controller_manager]: Available classes:
[ros2_control_node-4] [INFO] [1700919707.526075634] [controller_manager]: controller_manager/test_controller
[ros2_control_node-4] [INFO] [1700919707.526103543] [controller_manager]:
controller_manager/test_controller failed init
[ros2_control_node-4] [INFO] [1700919707.526129859] [controller_manager]:
controller_manager/test_controller_with_interfaces
[ros2_control_node-4] [INFO] [1700919707.526158527] [controller_manager]:
joint_state_broadcaster/JointStateBroadcaster
[ros2_control_node-4] [INFO] [1700919707.526212030] [controller_manager]:
joint_trajectory_controller/JointTrajectoryController
[ros2_control_node-4] [INFO] [1700919707.528960763] [controller_manager]:
controller_manager/test_chainable_controller
[spawner-6] [FATAL] [1700919707.533096584] [spawner_diff_cont]: Failed loading controller diff_cont
[spawner-5] [INFO] [1700919707.759520886] [spawner_joint_broad]: Loaded joint_broad
[ros2_control_node-4] [INFO] [1700919707.765827910] [controller_manager]: Configuring controller 'joint_broad'
[ros2_control_node-4] [INFO] [1700919707.766330214] [joint_broad]: 'joints' or 'interfaces' parameter is empty. All
available state interfaces will be published
[spawner-5] [INFO] [1700919707.954555349] [spawner_joint_broad]: Configured and activated joint_broad
[ERROR] [spawner-6]: process has died [pid 196020, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner
diff_cont --ros-args'].
[INFO] [spawner-5]: process has finished cleanly [pid 196018]
```

my initial thoughts that it dose not see the "my_controllers.yaml" file but it runs the "joint_broad" without any problems

**ali_ghasemi**

Nov '23

Hi Josh!

I'm planning to simulate an AUV for 3D underwater object tracking with camera sensor using ROS and Gazebo. Do you have Any recommendations on the ROS version, Gazebo version, and plugins for simulation this scenario? or GitHub directory?

**Agnes_Soesanto**

Nov '23

hello, i have problem with the diffdrive_arduino, it can't be compiled. im using the ubuntu 22.04 with ros humble and i can install the ros-foxy, so i use the ros-humble. but after i install the ros-humble-ros2-control & ros-humble-ros2-controllers the diffdrive_arduino still won't compiled. can u help me?? thankyou
agnes@raspberrypi:~/ros2_ws\$ colcon build --symlink-install
[8.185s] WARNING:colcon.colcon_core.package_selection:Some selected packages are already built in one or more underlay workspaces:
'rplidar_ros' is in: /home/agnes/ros2_ws/install/rplidar_ros
If a package in a merged underlay workspace is overridden and it installs headers, then all packages in the overlay must sort their include directories by workspace order. Failure to do so may result in build failures or undefined behavior at run time.
If the overridden package is used by another package in any underlay, then the overriding package in the overlay must be API and ABI compatible or undefined behavior at run time may occur.

If you understand the risks and want to override a package anyways, add the following to the command line:
--allow-overriding rplidar_ros

This may be promoted to an error in a future release of colcon-override-check.

Starting >>> serial

Starting >>> articubot_one

Starting >>> ldlidar_stl_ros2

Starting >>> rplidar_ros

Finished <<< articubot_one [3.33s]

Finished <<< ldlidar_stl_ros2 [3.37s]

Finished <<< serial [4.88s]

Starting >>> diffdrive_arduino

Finished <<< rplidar_ros [4.61s]

— stderr: diffdrive_arduino

In file included from /home/agnes/ros2_ws/src/diffdrive_arduino/src/fake_robot.cpp:1:

/home/agnes/ros2_ws/src/diffdrive_arduino/include/diffdrive_arduino/fake_robot.h:7:10:

fatal error: hardware_interface/base_interface.hpp: No such file or directory

7 | #include "hardware_interface/base_interface.hpp"

| ^~~~~~

compilation terminated.

gmake[2]: * [CMakeFiles/fake_robot.dir/build.make:76:**

CMakeFiles/fake_robot.dir/src/fake_robot.cpp.o] Error 1

gmake[1]: * [CMakeFiles/Makefile2:165: CMakeFiles/fake_robot.dir/all] Error 2**

gmake[1]: * Waiting for unfinished jobs...**

In file included from

/home/agnes/ros2_ws/src/diffdrive_arduino/src/diffdrive_arduino.cpp:1:

/home/agnes/ros2_ws/src/diffdrive_arduino/include/diffdrive_arduino/diffdrive_arduino.h:7:

10: fatal error: hardware_interface/base_interface.hpp: No such file or directory
7 | #include "hardware_interface/base_interface.hpp"
| ^~~~~~
compilation terminated.
gmake[2]: *** [CMakeFiles/diffdrive_arduino.dir/build.make:76:
CMakeFiles/diffdrive_arduino.dir/src/diffdrive_arduino.cpp.o] Error 1
gmake[1]: *** [CMakeFiles/Makefile2:139: CMakeFiles/diffdrive_arduino.dir/all] Error 2
gmake: *** [Makefile:146: all] Error 2

Failed <<< diffdrive_arduino [3.85s, exited with code 2]

Summary: 4 packages finished [14.0s]
1 package failed: diffdrive_arduino
1 package had stderr output: diffdrive_arduino

1 reply



Omar_Khaled

Dec '23

hello Mr joshNewans i have a problem when i try to run this command ros2 launch articubot_one launch_robot.launch.py
the error is :
[INFO] [launch]: All log files can be found below /home/omar/.ros/log/2023-12-19-14-15-50-067095-omarlegiony540-15500
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [15503]
[INFO] [twist_mux-2]: process started with pid [15505]
[robot_state_publisher-1] Parsing robot urdf xml string.
[robot_state_publisher-1] Link base_footprint had 0 children
[robot_state_publisher-1] Link chassis had 3 children
[robot_state_publisher-1] Link camera_link had 1 children
[robot_state_publisher-1] Link camera_link_optical had 0 children
[robot_state_publisher-1] Link caster_wheel had 0 children
[robot_state_publisher-1] Link laser_frame had 0 children
[robot_state_publisher-1] Link left_wheel had 0 children
[robot_state_publisher-1] Link right_wheel had 0 children
[robot_state_publisher-1] [INFO] [1702988150.253367756] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1702988150.253457840] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1702988150.253472042] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-1] [INFO] [1702988150.253482521] [robot_state_publisher]: got segment camera_link_optical
[robot_state_publisher-1] [INFO] [1702988150.253492593] [robot_state_publisher]: got segment caster_wheel
[robot_state_publisher-1] [INFO] [1702988150.253502422] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1702988150.253512065] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1702988150.253521801] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1702988150.253531529] [robot_state_publisher]: got segment right_wheel
[twist_mux-2] [INFO] [1702988150.253417560] [twist_mux]: Topic handler 'topics.joystick' subscribed to topic 'cmd_vel_joy': timeout = 0.500000s ,
priority = 100.
[twist_mux-2] [INFO] [1702988150.253767902] [twist_mux]: Topic handler 'topics.navigation' subscribed to topic 'cmd_vel': timeout = 0.500000s ,
priority = 10.
[twist_mux-2] [INFO] [1702988150.253950440] [twist_mux]: Topic handler 'topics.tracker' subscribed to topic 'cmd_vel_tracker': timeout =
0.500000s , priority = 20.
[INFO] [ros2_control_node-3]: process started with pid [15541]
[INFO] [spawner.py-4]: process started with pid [15543]
[INFO] [spawner.py-5]: process started with pid [15545]
[ros2_control_node-3] terminate called after throwing an instance of 'pluginlib::LibraryLoadException'
[ros2_control_node-3] what(): According to the loaded plugin descriptions the class diffdrive_arduino/DiffDriveArduino with base class type
hardware_interface::SystemInterface does not exist. Declared types are fake_components/GenericSystem
test_hardware_components/TestSystemCommandModesTest_hardware_components/TestTwoJointSystem test_system
turtlebot3_manipulation_hardware/TurtleBot3ManipulationSystemHardware
[spawner.py-4] [INFO] [1702988154.738035052] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1702988154.742421839] [spawner_diff_cont]: Waiting for /controller_manager services
[ERROR] [ros2_control_node-3]: process has died [pid 15541, exit code -6, cmd '/opt/ros/foxy/lib/controller_manager/ros2_control_node --ros-args
--params-file /tmp/launch_params_i511tc0_--params-file
/home/omar/colcon_ws/install/articubot_one/share/articubot_one/config/my_controllers.yaml'].
[spawner.py-4] [INFO] [1702988156.754901118] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1702988156.758037875] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-4] [INFO] [1702988158.770261931] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1702988158.773408347] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-4] [INFO] [1702988160.788893056] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1702988160.791728848] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-4] [INFO] [1702988162.807066439] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1702988162.811132170] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-4] [ERROR] [1702988164.823391319] [spawner_joint_broad]: Controller manager not available
[spawner.py-5] [ERROR] [1702988164.828567043] [spawner_diff_cont]: Controller manager not available
[ERROR] [spawner.py-4]: process has died [pid 15543, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py joint_broad --ros-args'].
[ERROR] [spawner.py-5]: process has died [pid 15545, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py diff_cont --ros-args'].
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[twist_mux-2] [INFO] [1702988175.110388053] [rclcpp]: signal_handler(signal_value=2)
[robot_state_publisher-1] [INFO] [1702988175.110824006] [rclcpp]: signal_handler(signal_value=2)
[INFO] [robot_state_publisher-1]: process has finished cleanly [pid 15503]
[INFO] [twist_mux-2]: process has finished cleanly [pid 15505]



Aboude_zoua

Dec '23

Hi Josh ,
I'm having the same issue like others with controller manager some how is not working .
I've been stuck in this for 4 days now with no luck ! any chance you recommend something ? I echo /cmd_vel_unstamped in robot and I can see
the values passing
[spawner-4] [INFO] [1703827838.665064167] [spawner_joint_broad]: Waiting for '/controller_manager' node to exist
[spawner-5] [INFO] [1703827838.785338511] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[spawner-4] [INFO] [1703827840.711671532] [spawner_joint_broad]: Waiting for '/controller_manager' node to exist
[spawner-5] [INFO] [1703827840.828404607] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[spawner-4] [INFO] [1703827842.760930707] [spawner_joint_broad]: Waiting for '/controller_manager' node to exist
[spawner-5] [INFO] [1703827842.877129094] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[spawner-4] [INFO] [1703827844.805368341] [spawner_joint_broad]: Waiting for '/controller_manager' node to exist
[spawner-5] [INFO] [1703827844.923151119] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[spawner-4] [ERROR] [1703827846.646826040] [spawner_joint_broad]: Controller manager not available
[spawner-5] [INFO] [1703827846.763479498] [spawner_diff_cont]: Controller manager not available
[ERROR] [spawner-4]: process has died [pid 7297, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner joint_broad --ros-args'].
[ERROR] [spawner-5]: process has died [pid 7299, exit code 1, cmd '/opt/ros/humble/lib/controller_manager/spawner diff_cont --ros-args'].

Also I ran controller_manager and no connection
ros2 run controller_manager spawner diff_cont
[INFO] [1703828709.701625640] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[INFO] [1703828711.722023739] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[INFO] [1703828713.742430911] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[INFO] [1703828715.764174999] [spawner_diff_cont]: Waiting for '/controller_manager' node to exist
[ERROR] [1703828717.784713751] [spawner_diff_cont]: Controller manager not available
[ros2run]: Process exited with failure 1

1 reply



Aboude_zoua

► Agnes_Soesanto Dec '23



Omar_Khaled

JoshNewans Dec '23

Hi Josh,

i am trying to launch launch_robot.launch.py but in my dev machine to try the motors with teleop and i got this problem when launch it (Note i donot have diffarduinio_drive in my packages and when i try to have it and build it , it give me error)

```
[INFO] [launch]: All log files can be found below /home/omar/.ros/log/2023-12-30-21-04-24-680391-omar/legion540-26297
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [26300]
[INFO] [twist_mux-2]: process started with pid [26302]
[twist_mux-2] [INFO] [1703963064.968940347] [twist_mux]: Topic handler 'topics.joystick' subscribed to topic 'cmd_vel_joy': timeout = 0.500000s , priority = 100.
[twist_mux-2] [INFO] [1703963064.969531345] [twist_mux]: Topic handler 'topics.navigation' subscribed to topic 'cmd_vel': timeout = 0.500000s , priority = 10.
[robot_state_publisher-1] Parsing robot urdf xml string.
[robot_state_publisher-1] Link base_footprint had 0 children
[robot_state_publisher-1] Link chassis had 2 children
[robot_state_publisher-1] Link camera_link had 1 children
[robot_state_publisher-1] Link camera_link_optical had 0 children
[robot_state_publisher-1] Link laser_frame had 0 children
[robot_state_publisher-1] Link left_wheel had 0 children
[robot_state_publisher-1] Link leftb_wheel had 0 children
[robot_state_publisher-1] Link leftf_wheel had 0 children
[robot_state_publisher-1] Link right_wheel had 0 children
[robot_state_publisher-1] Link rightb_wheel had 0 children
[robot_state_publisher-1] Link rightf_wheel had 0 children
[robot_state_publisher-1] [INFO] [1703963064.974543580] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1703963064.974624726] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1703963064.974637111] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-1] [INFO] [1703963064.974645299] [robot_state_publisher]: got segment camera_link_optical
[robot_state_publisher-1] [INFO] [1703963064.974652976] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1703963064.974660158] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1703963064.974666931] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1703963064.974673816] [robot_state_publisher]: got segment leftb_wheel
[robot_state_publisher-1] [INFO] [1703963064.974680677] [robot_state_publisher]: got segment leftf_wheel
[robot_state_publisher-1] [INFO] [1703963064.974687395] [robot_state_publisher]: got segment right_wheel
[robot_state_publisher-1] [INFO] [1703963064.974694133] [robot_state_publisher]: got segment rightb_wheel
[robot_state_publisher-1] [INFO] [1703963064.974701012] [robot_state_publisher]: got segment rightf_wheel
[INFO] [ros2_control_node-3]: process started with pid [26338]
[INFO] [spawner.py-4]: process started with pid [26340]
[INFO] [spawner.py-5]: process started with pid [26342]
[ros2_control_node-3] terminate called after throwing an instance of 'pluginlib::LibraryLoadException'
[ros2_control_node-3] what(): According to the loaded plugin descriptions the class diffdrive_arduino/DiffDriveArduino with base class type hardware_interface::SystemInterface does not exist. Declared types are fake_components/GenericSystem
testHardwareComponents/TestSystemCommandModes testHardwareComponents/TestTwoJointSystem testSystem
[spawner.py-4] [INFO] [1703963069.430926611] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1703963069.431065690] [spawner_diff_cont]: Waiting for /controller_manager services
[ERROR] [ros2_control_node-3]: process has died [pid 26338, exit code -6, cmd '/opt/ros/foxy/lib/controller_manager/ros2_control_node --ros-args --params-file /tmp/launch_params_8fujlyhu --params-file /home/omar/real_ws/install/medo/share/medo/config/my_controllers.yaml'].
[spawner.py-4] [INFO] [1703963071.446737365] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1703963071.447554445] [spawner_diff_cont]: Waiting for /controller_manager services
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[twist_mux-2] [INFO] [1703963073.455811200] [rcicpp]: signal_handler(signal_value=2)
[robot_state_publisher-1] [INFO] [1703963073.455834934] [rcicpp]: signal_handler(signal_value=2)
[spawner.py-5] Traceback (most recent call last):
[spawner.py-5] File "/opt/ros/foxy/lib/controller_manager/spawner.py", line 204, in
[spawner.py-5] sys.exit(main())
[spawner.py-5] File "/opt/ros/foxy/lib/controller_manager/spawner.py", line 118, in main
[spawner.py-5] if not wait_for_controller_manager(node, controller_manager_name,
[spawner.py-5] File "/opt/ros/foxy/lib/controller_manager/spawner.py", line 59, in wait_for_controller_manager
[spawner.py-5] time.sleep(0.2)
[spawner.py-5] KeyboardInterrupt
[spawner.py-4] Traceback (most recent call last):
[spawner.py-4] File "/opt/ros/foxy/lib/controller_manager/spawner.py", line 204, in
[spawner.py-4] sys.exit(main())
[spawner.py-4] File "/opt/ros/foxy/lib/controller_manager/spawner.py", line 118, in main
[spawner.py-4] if not wait_for_controller_manager(node, controller_manager_name,
[spawner.py-4] File "/opt/ros/foxy/lib/controller_manager/spawner.py", line 59, in wait_for_controller_manager
[spawner.py-4] time.sleep(0.2)
[spawner.py-4] KeyboardInterrupt
[INFO] [twist_mux-2]: process has finished cleanly [pid 26302]
[INFO] [robot_state_publisher-1]: process has finished cleanly [pid 26300]
[ERROR] [spawner.py-5]: process has died [pid 26342, exit code -2, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py diff_cont --ros-args'].
[ERROR] [spawner.py-4]: process has died [pid 26340, exit code -2, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py joint_broad --ros-args'].
```



Aboude_zoua

Aboude_zoua Dec '23

I was able to resolve this issue by cloning the repo from buzzology github and then rebuilt the whole project with colcon .



omar_salah

Jan '24

hey Josh, thanks for the amazing tutorial and the generous content in general i see you have commented on this issue before but i can't seem to figure it out yet it happens on both the machine and the pi

```
[spawner.py-6] [INFO] [1704429643.856203298] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1704429643.945700660] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-6] [INFO] [1704429645.866817026] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1704429645.955195869] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-6] [INFO] [1704429647.880334317] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1704429647.967992784] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-6] [INFO] [1704429649.896868913] [spawner_joint_broad]: Waiting for /controller_manager services
[spawner.py-5] [INFO] [1704429649.984468641] [spawner_diff_cont]: Waiting for /controller_manager services
[spawner.py-6] [ERROR] [1704429651.911524114] [spawner_joint_broad]: Controller manager not available
[ERROR] [spawner.py-6]: process has died [pid 9922, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py joint_broad --ros-args'].
[spawner.py-5] [ERROR] [1704429651.999166283] [spawner_diff_cont]: Controller manager not available
[ERROR] [spawner.py-5]: process has died [pid 9920, exit code 1, cmd '/opt/ros/foxy/lib/controller_manager/spawner.py diff_cont --ros-args'].
```



jonnyr

Douglas_Beach Jan '24

See the documentation on the hardware_interface for the ros2_control, on github

Migration from Foxy to newer versions

Between Foxy and Galactic we made substantial changes to the interface of hardware components to enable management of their lifecycle. The following list shows a set of quick changes to port existing hardware components to Galactic:

1. Rename configure to on_init and change return type to CallbackReturn
2. If using BaseInterface as base class then you should remove it. Specifically, change:

```
hardware_interface::BaseInterface<hardware_interface::[Actuator|Sensor|System]Interface>
```

to

- Remove include of headers base_interface.hpp and hardware_interface_status_values.hpp

**martoli0d**

► Afiq003 Jan '24

Hey [@Afiq003](#),
i got the same error... but as [@JoshNewans](#):

i'm a bit confused by your solution.

what exactly did u do to resolve this? Thanks alot in advance!

EDIT: found the issue. in `rsp.launch.py` i needed to insert an blank before the "sim_mode" parameter as follows:

old:

```
robot_description_config = Command(['xacro ', xacro_file, ' use_ros2_control:=', use_ros2_control, 'sim_mode:=', use_sim_time])
```

new:

```
robot_description_config = Command(['xacro ', xacro_file, ' use_ros2_control:=', use_ros2_control, ' sim_mode:=', use_sim_time])
```

**KaushikBoora**

Mar '24

Hi [@JoshNewans](#) thanks a lot for the videos, They helped me a lot to understand ROS2 and how to build a differential drive bot. I am working on a 4 WD and have been following your tutorials until " [Using ros2_control to drive our robot (off the edge of the bench...)] (https://www.youtube.com/watch?v=4VVrTCnxv5w&t=601s)" video. I am facing some trouble after changing to ros2_control, The issue is that my robot is not moving either left or right in gazebo but it seems to move in Rviz as expected. I have searched all the online resources but did not find anything that helps.
here is the link to my repo [GitHub - BooraKaushik/autobot](#)
Video of what is happening, [IMG_5920.mov - Google Drive](#)
as you can see it is turning fine in rviz but not in gazebo.

I would really appreciate any help.
Thanks,
Kaushik Boora

**Hamza_Anwar**

Mar '24

Hi guys. I am new to ROS2 and I am following [@JoshNewans](#) tutorial to make my own robot. I am using an arduino uno board with polulu motors. I was able to make the robot work in gazebo/rviz but when I downloaded the diffdrive controller from Josh' github and started to port it on the robot using RPI 3b+, I get the error:

```
ros2 launch adrover_ROS2 launch_robot.launch.py
[INFO] [launch]: All log files can be found below /home/adrover/.ros/log/2024-03-20-14-58-26-819462-adrover-2942
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [2945]
[robot_state_publisher-1] Parsing robot urdf xml string.
[robot_state_publisher-1] Link chassis_link_back had 1 children
[robot_state_publisher-1] Link laser_frame had 0 children
[robot_state_publisher-1] Link chassis_link_front had 1 children
[robot_state_publisher-1] Link lcd_link had 1 children
[robot_state_publisher-1] Link camera_link had 1 children
[robot_state_publisher-1] Link camera_link_optical had 0 children
[robot_state_publisher-1] Link chassis_link_left had 2 children
[robot_state_publisher-1] Link caster_wheel_link_lb had 0 children
[robot_state_publisher-1] Link caster_wheel_link_lf had 0 children
[robot_state_publisher-1] Link chassis_link_right had 2 children
[robot_state_publisher-1] Link caster_wheel_link_rb had 0 children
[robot_state_publisher-1] Link caster_wheel_link_rf had 0 children
[robot_state_publisher-1] Link left_wheel_link had 0 children
[robot_state_publisher-1] Link right_wheel_link had 0 children
[robot_state_publisher-1] [INFO] [1710928708.453907263] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1710928708.454549749] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-1] [INFO] [1710928708.454657611] [robot_state_publisher]: got segment camera_link_optical
[robot_state_publisher-1] [INFO] [1710928708.454703444] [robot_state_publisher]: got segment caster_wheel_link_lb
[robot_state_publisher-1] [INFO] [1710928708.454747245] [robot_state_publisher]: got segment caster_wheel_link_lf
[robot_state_publisher-1] [INFO] [1710928708.454788963] [robot_state_publisher]: got segment caster_wheel_link_rb
[robot_state_publisher-1] [INFO] [1710928708.454830316] [robot_state_publisher]: got segment caster_wheel_link_rf
[robot_state_publisher-1] [INFO] [1710928708.454871253] [robot_state_publisher]: got segment chassis_link_back
[robot_state_publisher-1] [INFO] [1710928708.454913543] [robot_state_publisher]: got segment chassis_link_front
[robot_state_publisher-1] [INFO] [1710928708.454954636] [robot_state_publisher]: got segment chassis_link_left
[robot_state_publisher-1] [INFO] [1710928708.454996562] [robot_state_publisher]: got segment chassis_link_right
[robot_state_publisher-1] [INFO] [1710928708.455047186] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1710928708.455090831] [robot_state_publisher]: got segment lcd_link
[robot_state_publisher-1] [INFO] [1710928708.455134007] [robot_state_publisher]: got segment left_wheel_link
[robot_state_publisher-1] [INFO] [1710928708.455264005] [robot_state_publisher]: got segment right_wheel_link
[INFO] [ros2_control_node-2]: process started with pid [2957]
[ros2_control_node-2] terminate called after throwing an instance of 'std::runtime_error'
[ros2_control_node-2] what(): invalid URDF passed in to robot parser
[ERROR] [ros2_control_node-2]: process has died [pid 2957, exit code -6, cmd
'/opt/ros/foxy/lib/controller_manager/ros2_control_node --ros-args --params-file /tmp/launch_params_y8g2xwtv --params-file /home/adrover/adrover_ws/install/adrover_ROS2/share/adrover_ROS2/config/adrover_controller.yaml'].
```

I have commented out the spawner in my launch files since the controller manager terminates automatically. Also my description file looks like:

```
<?xml version="1.0"?>
<robot xmlns:xacro="http://www.ros.org/wiki/xacro">

  <ros2_control name="RealRobot" type="system">

    <hardware>
      <plugin>diffdrive_arduino/DiffDriveArduino</plugin>
      <param name="left_wheel_names">left_wheel_joint</param>
      <param name="right_wheel_names">right_wheel_joint</param>
      <param name="loop_rate">30</param>
      <param name="device">/dev/serial/by-id/usb-1a86_USB2.0-Ser-_if00-port0</param>
      <param name="baud_rate">57600</param>
      <param name="timeout">1000</param>
      <param name="enc_counts_per_rev">8400</param>
    </hardware>

    <joint name="left_wheel_joint">
      <command_interface name="velocity">
        <param name="min">-10</param>
        <param name="max">10</param>
      </command_interface>
      <state_interface name="velocity" />
      <state_interface name="position" />
    </joint>
  </ros2_control>
</robot>
```

```
</joint>

<joint name="right_wheel_joint">
  <command_interface name="velocity">
    <param name="min">-10</param>
    <param name="max">10</param>
  </command_interface>
  <state_interface name="velocity" />
  <state_interface name="position" />
</joint>

</ros2_control>
<gazebo>
  <plugin name="gazebo_ros2_control" filename="libgazebo_ros2_control.so">
    <parameters>$(find adrover_R0S2)/config/adrover_controller.yaml</parameters>
  </plugin>
</gazebo>

</robot>
```

Kindly any help in this regard?

1 reply



UNAIB_USMANI

Mar '24

hi , i m an issue while launch ros2 control list hardware interface output will coming like this could not contact to service controller manager
can anyone please help me out



Hamza_Anwar

► Hamza_Anwar Mar '24

It worked for me. For reference, I switched to ROS foxy but the issue persisted. I was connected to RPi over ssh and using nano for editing. I ultimately connected to vscode over remote connection, replaced all my code with the snippets provided in the blog by Josh (I did this to ros2_control.xacro and the launch_robot.launch.py files though I couldn't see any difference between the two), removed the gazebo_control part completely instead of commenting it out and recompiled and run the launch file. It worked... Thankyou everyone and specially @JoshNewans for these great tutorials.



Lakess

Apr '24

Hi Josh

I'm trying to run "ros2 launch articubot_one launch_robot.launch.py" but get this error, I wonder if you can guide me to a solution. Thanks

```
[INFO] [launch]: All log files can be found below /home/diaz/.ros/log/2024-04-04-06-44-23-550460-diaz-desktop-13685
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [13688]
[robot_state_publisher-1] 1712227464.651032 [0] robot_stat: using network interface wlan0 (udp/192.168.1.107) selected arbitrarily from: wlan0,
docker0
[robot_state_publisher-1] Parsing robot urdf xml string.
[robot_state_publisher-1] Link chassis had 4 children
[robot_state_publisher-1] Link camera_link had 1 children
[robot_state_publisher-1] Link camera_link_optical had 0 children
[robot_state_publisher-1] Link caster_wheel had 0 children
[robot_state_publisher-1] Link face_link had 0 children
[robot_state_publisher-1] Link laser_frame had 0 children
[robot_state_publisher-1] Link left_wheel had 0 children
[robot_state_publisher-1] Link right_wheel had 0 children
[robot_state_publisher-1] [INFO] [1712227464.679112441] [robot_state_publisher]: got segment base_link
[robot_state_publisher-1] [INFO] [1712227464.679405597] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-1] [INFO] [1712227464.679503966] [robot_state_publisher]: got segment camera_link_optical
[robot_state_publisher-1] [INFO] [1712227464.679596512] [robot_state_publisher]: got segment caster_wheel
[robot_state_publisher-1] [INFO] [1712227464.679661025] [robot_state_publisher]: got segment chassis
[robot_state_publisher-1] [INFO] [1712227464.679858499] [robot_state_publisher]: got segment face_link
[robot_state_publisher-1] [INFO] [1712227464.679919908] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-1] [INFO] [1712227464.679992325] [robot_state_publisher]: got segment left_wheel
[robot_state_publisher-1] [INFO] [1712227464.680050182] [robot_state_publisher]: got segment right_wheel
Task exception was never retrieved
future: <Task finished name='Task-12' coro=<LaunchService._process_one_event() done, defined at /opt/ros/foxy/lib/python3.8/site-
packages/launch/launch_service.py:226> exception=SubstitutionFailure('executed command showed stderr output. Command: ros2 param get --
hide-type /robot_state_publisher robot_description\nCaptured stderr output:\n1712227469.524891 [0] ros2: using network interface wlan0
(udp/192.168.1.107) selected arbitrarily from: wlan0, docker0\n1712227470.409871 [0] ros2: using network interface wlan0 (udp/192.168.1.107)
selected arbitrarily from: wlan0, docker0\n')>
Traceback (most recent call last):
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 228, in _process_one_event
await self._process_event(next_event)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_service.py", line 248, in _process_event
visit_all_entities_and_collect_futures(entity, self._context))
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py", line 45, in
visit_all_entities_and_collect_futures
futures_to_return += visit_all_entities_and_collect_futures(sub_entity, context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/visit_all_entities_and_collect_futures_impl.py", line 38, in
visit_all_entities_and_collect_futures
sub_entities = entity.visit(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/action.py", line 108, in visit
return self.execute(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/actions/node.py", line 444, in execute
self._perform_substitutions(context)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/actions/node.py", line 399, in _perform_substitutions
evaluated_parameters = evaluate_parameters(context, self._parameters)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/utilities/evaluate_parameters.py", line 164, in evaluate_parameters
output_params.append(evaluate_parameter_dict(context, param))
File "/opt/ros/foxy/lib/python3.8/site-packages/launch_ros/utilities/evaluate_parameters.py", line 71, in evaluate_parameter_dict
evaluated_value = perform_substitutions(context, list(value))
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/perform_substitutions_impl.py", line 26, in perform_substitutions
return ''.join([context.perform_substitution(sub) for sub in subs])
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/utilities/perform_substitutions_impl.py", line 26, in
return ''.join([context.perform_substitution(sub) for sub in subs])
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/launch_context.py", line 232, in perform_substitution
return substitution.perform(self)
File "/opt/ros/foxy/lib/python3.8/site-packages/launch/substitutions/command.py", line 122, in perform
raise SubstitutionFailure(on_stderr_message)
launch.substitutions.substitution_failure.SubstitutionFailure: executed command showed stderr output. Command: ros2 param get --hide-type
/robot_state_publisher robot_description
Captured stderr output:
1712227469.524891 [0] ros2: using network interface wlan0 (udp/192.168.1.107) selected arbitrarily from: wlan0, docker0
1712227470.409871 [0] ros2: using network interface wlan0 (udp/192.168.1.107) selected arbitrarily from: wlan0, docker0

^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[robot_state_publisher-1] [INFO] [1712228830.109878910] [rclcpp]: signal_handler(signal_value=2)
[INFO] [robot_state_publisher-1]: process has finished cleanly [pid 13688]
```



glassTransition

May '24

I've been banging on this for some time and would appreciate any help that someone can offer. I've followed the update instructions for Humble to

the best of my ability and can run everything in the simulated environment. The error comes when I try to run on the real robot. I get this error:

```
[ros2_control_node-3] [WARN] [1717166914.463673109] [controller_manager]: [Deprecated] Passing the robot description parameter directly to the control_manager node is deprecated. Use '~/robot_description' topic from 'robot_state_publisher' instead.  
[ros2_control_node-3] [INFO] [1717168386.815847187] [resource_manager]: Loading hardware 'RealRobot'  
[ros2_control_node-3] terminate called after throwing an instance of 'pluginlib::LibraryLoadException'  
[ros2_control_node-3] what(): According to the loaded plugin descriptions the class diffdrive_arduino/DiffDriveArduinoHardware with base class type hardware_interface::SystemInterface does not exist. Declared types are diffdrive_arduino/DiffDriveArduino fake_components/GenericSystem fake_robot/FakeRobot mock_components/GenericSystem testHardware_components/TestSystemCommandModes testHardware_components/TestTwoJointSystem
```

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